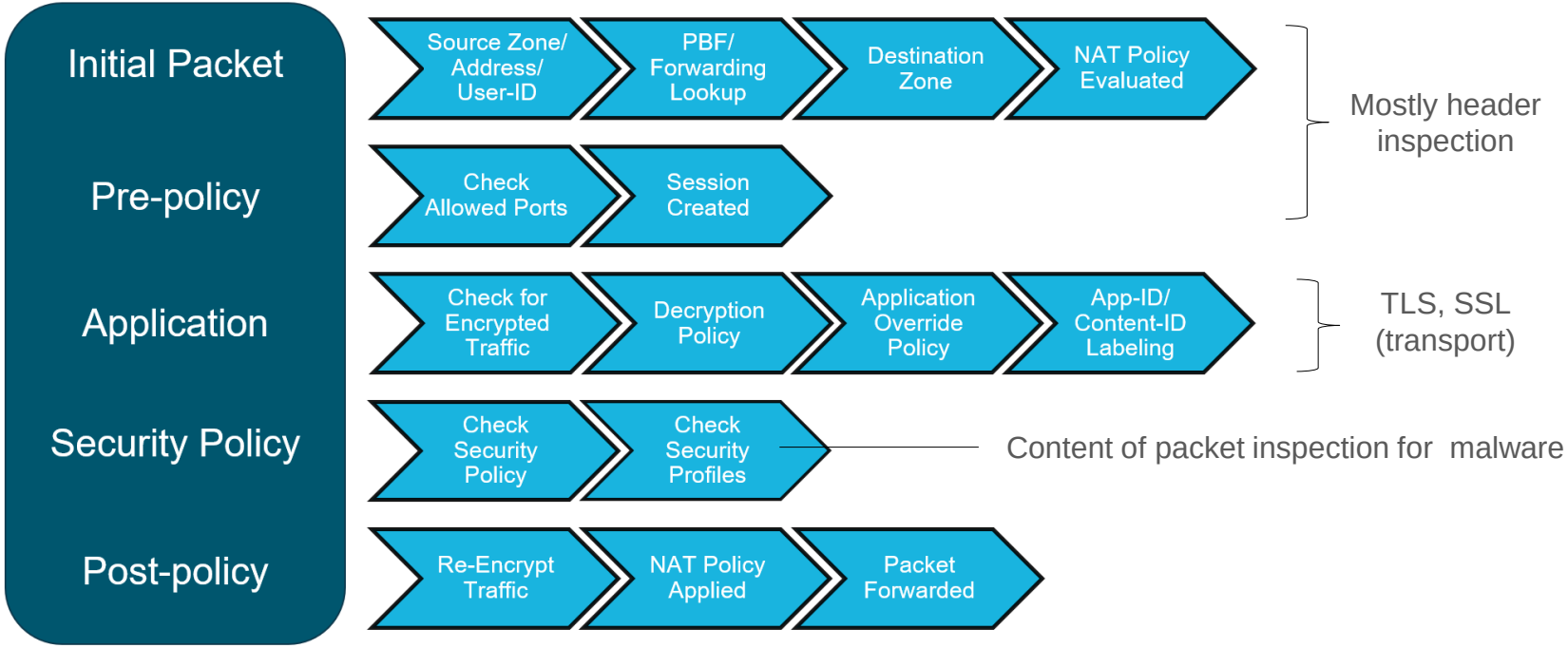
Next Generation Firewalls
Zones - Interface Configuration

Topics

- Security zones and interfaces
- Tap interfaces
- Virtual Wire interfaces
- Layer 2 interfaces
- Layer 3 interfaces
- Virtual routers
- VLAN interfaces
- Loopback interfaces
- Policy-based forwarding

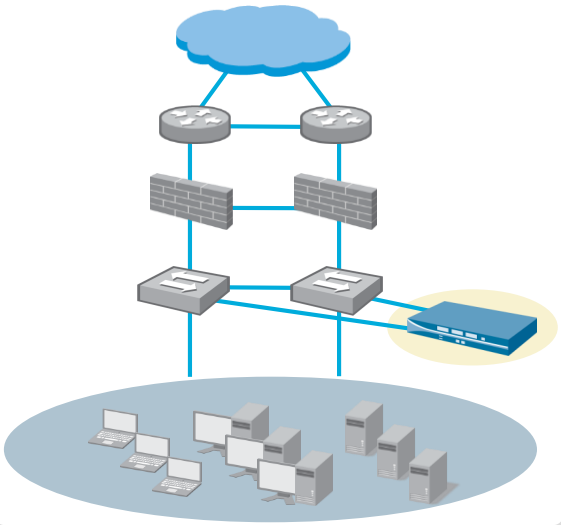
Flow Logic of the Next-Generation Firewall

- A simplified version of the flow logic of a packet traveling through a NGFW



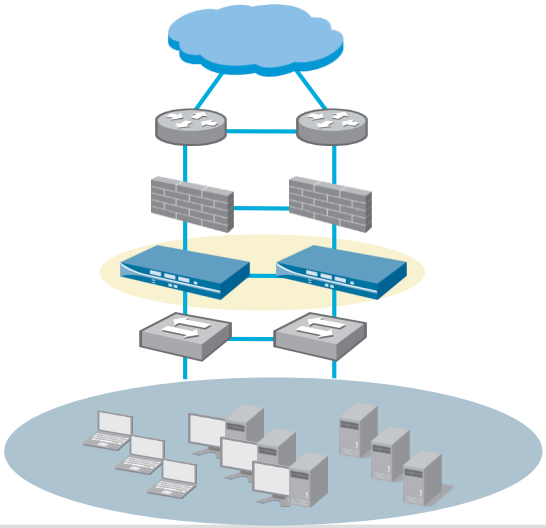
Deployment Options for Ethernet Interfaces

Tap



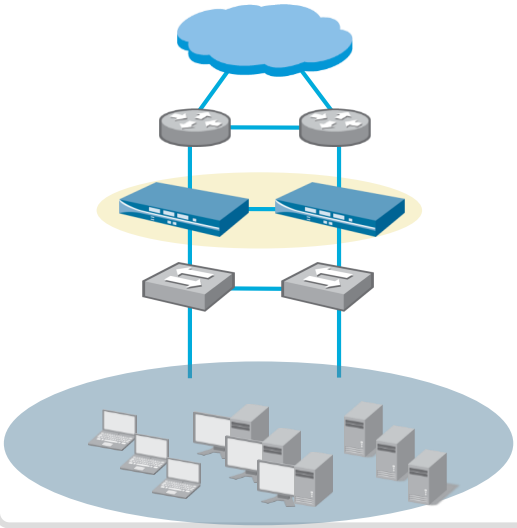
- Application, user, and content visibility without inline deployment
- Evaluation and audit of existing networks

Virtual Wire



- App-ID, Content-ID, User-ID, and SSL decryption
- Includes NAT capability
- No network addresses (IPs)
- No redesign of network topology

Layer 3

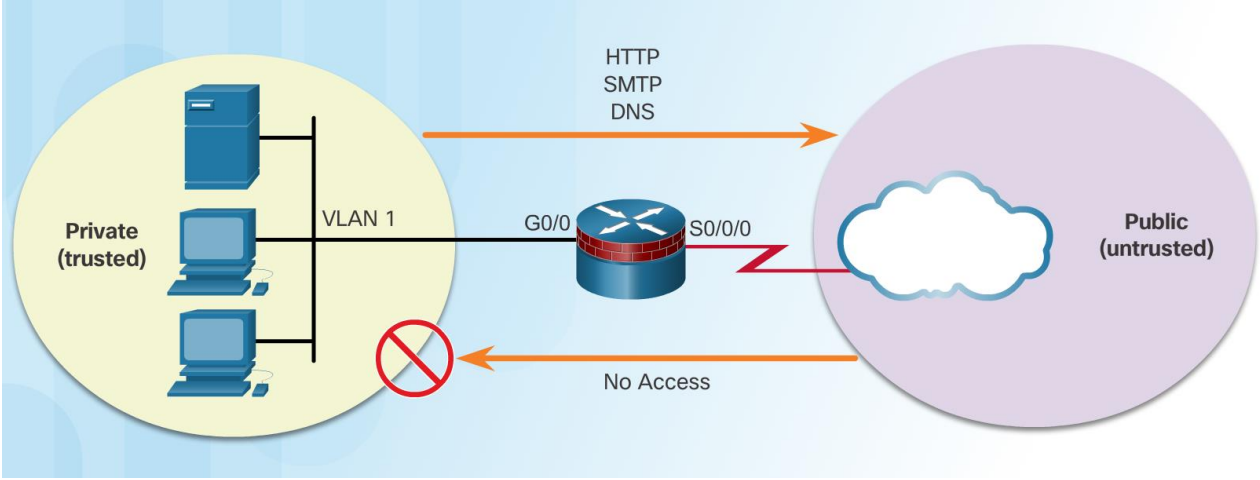


- All of the Virtual Wire mode capabilities
- Layer 3 services: virtual routers, VPN, routing protocols

Security Zones and Interfaces

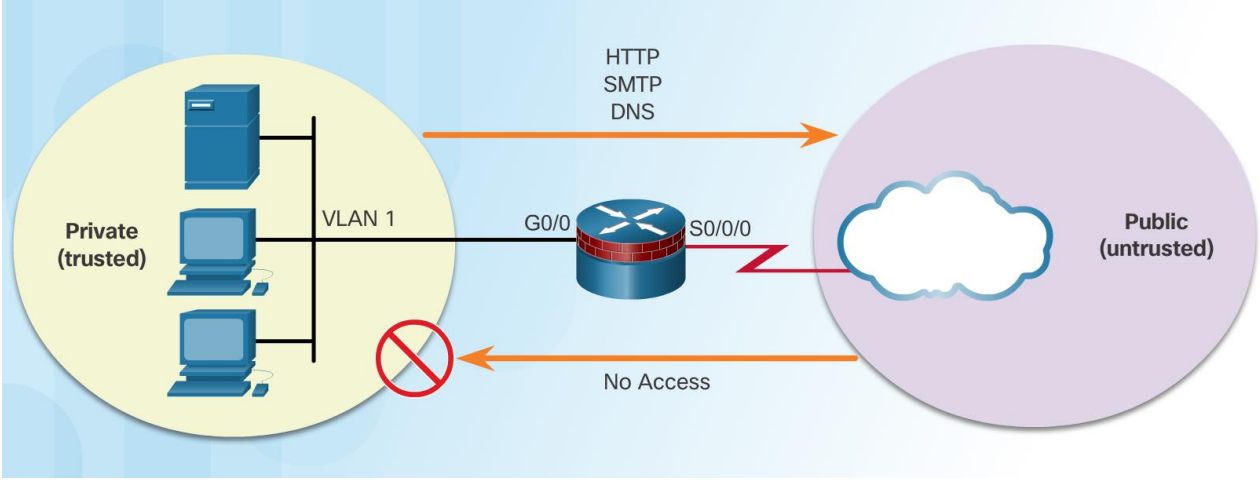
Security Zones

- Firewall design is primarily about device interfaces permitting or denying traffic based on source, destination, application type, and other information.
- Some designs are simple: one outside network and one inside network.
- Public (outside) network is untrusted, and private (inside) network is trusted.



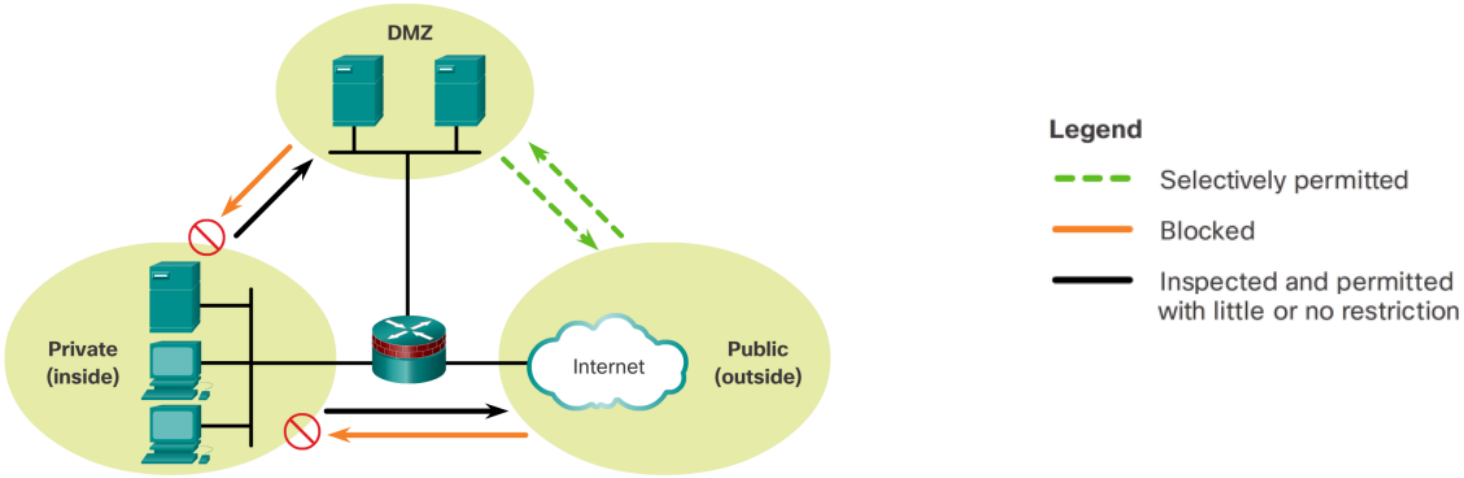
Security Zones

- Typically a firewall with two interfaces is configured as follows:
  - Traffic originating from private network is permitted; *returning traffic* from public network and associated with traffic that originated from private network is permitted.
  - Traffic originating from the public network and traveling to the private network is generally blocked.



Demilitarized Zone

- A demilitarized zone (DMZ) is a portion of a network bounded by a firewall or set of firewalls.
- Originally used as a military description for an area between military powers where conflict is not permitted.



Zone-Based Security Policies

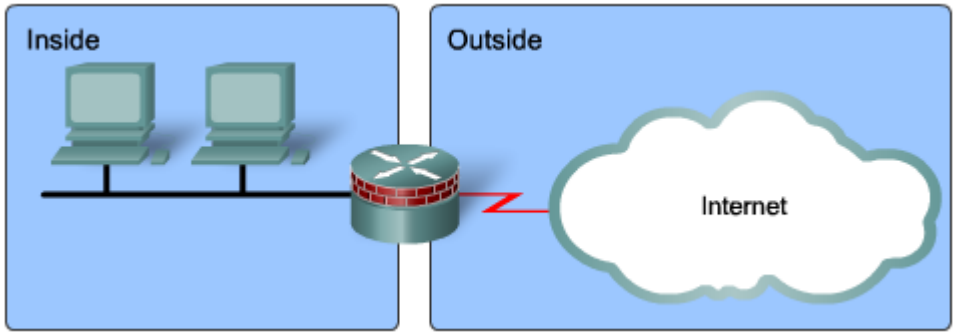
- Introduced in mid-late 2000s.
- Interfaces are assigned to zones and then an inspection policy is applied to traffic moving between the zones.
- The default policy is to block all traffic unless explicitly allowed.

Advantages of Zone-Based Security Policies

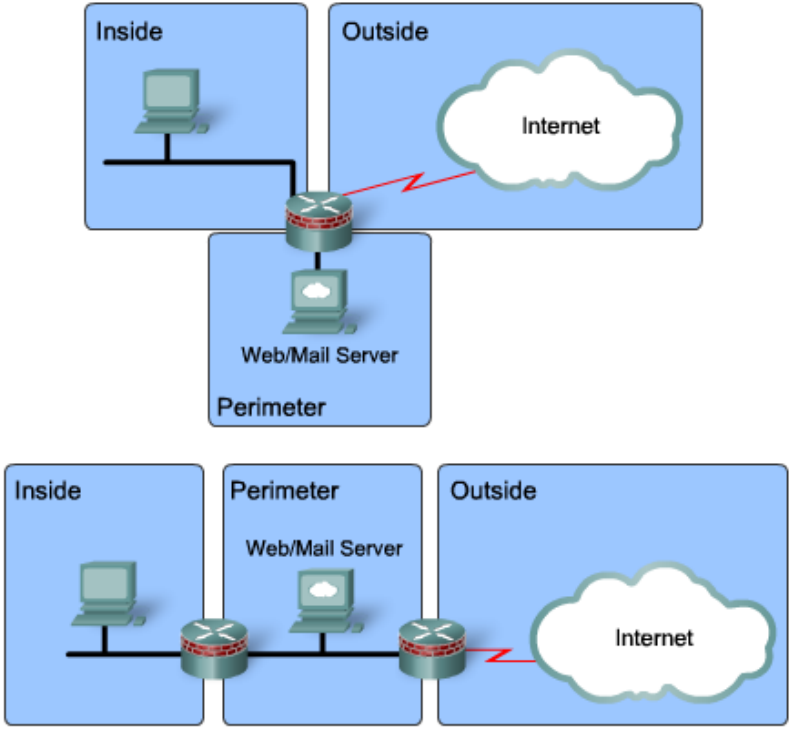
- Not dependent on access-control lists.
- The firewall security posture is to block unless explicitly allowed.
- Policies are easier to read and troubleshoot.
- Provides scalability; one policy affects any given traffic, instead of needing multiple policies and inspection actions.

Common Designs

- Typical designs



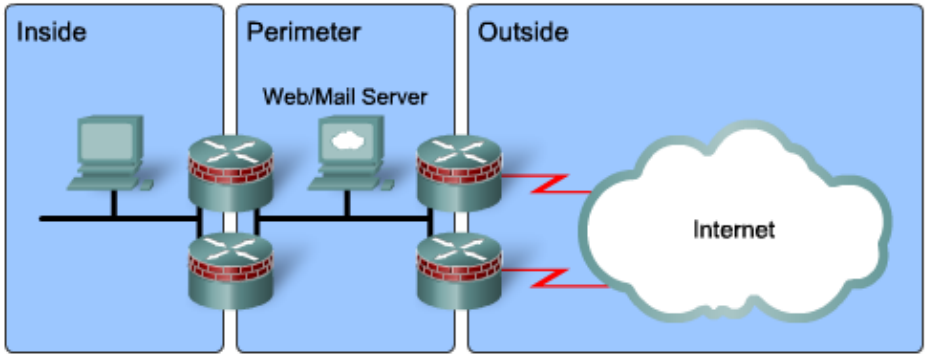
Two-zone design



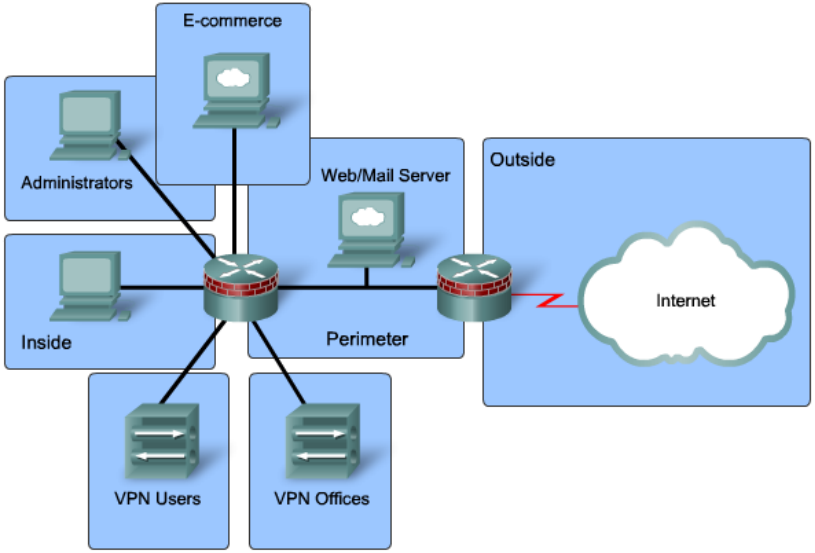
Three-zone design, with DMZ

Common Designs

- Typical designs



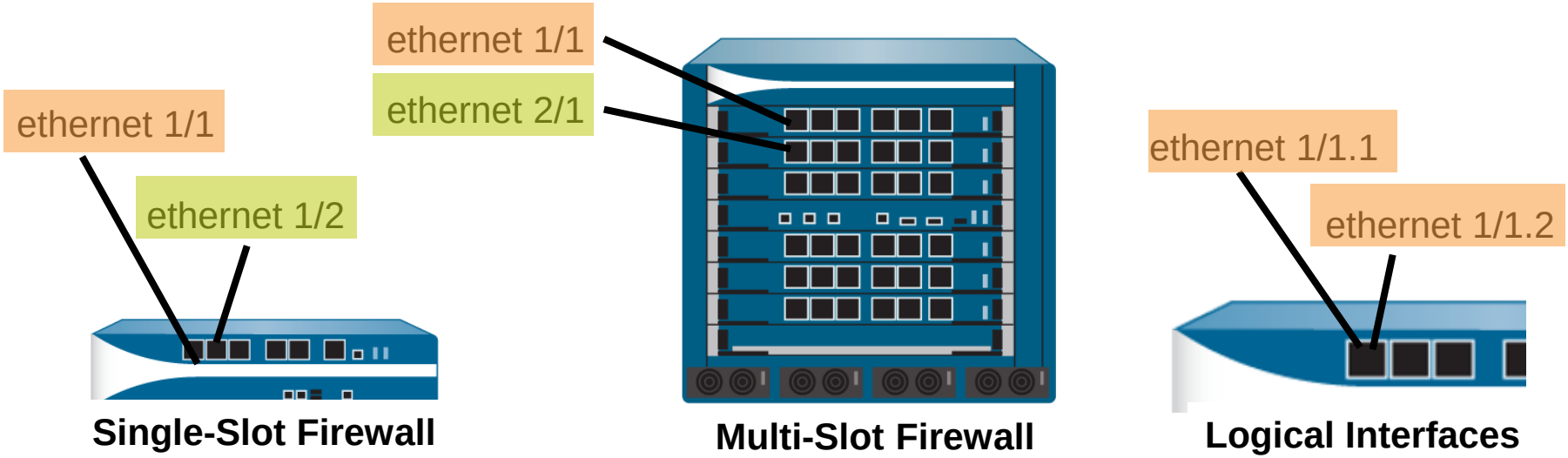
Redundant firewalls



Complex firewalls

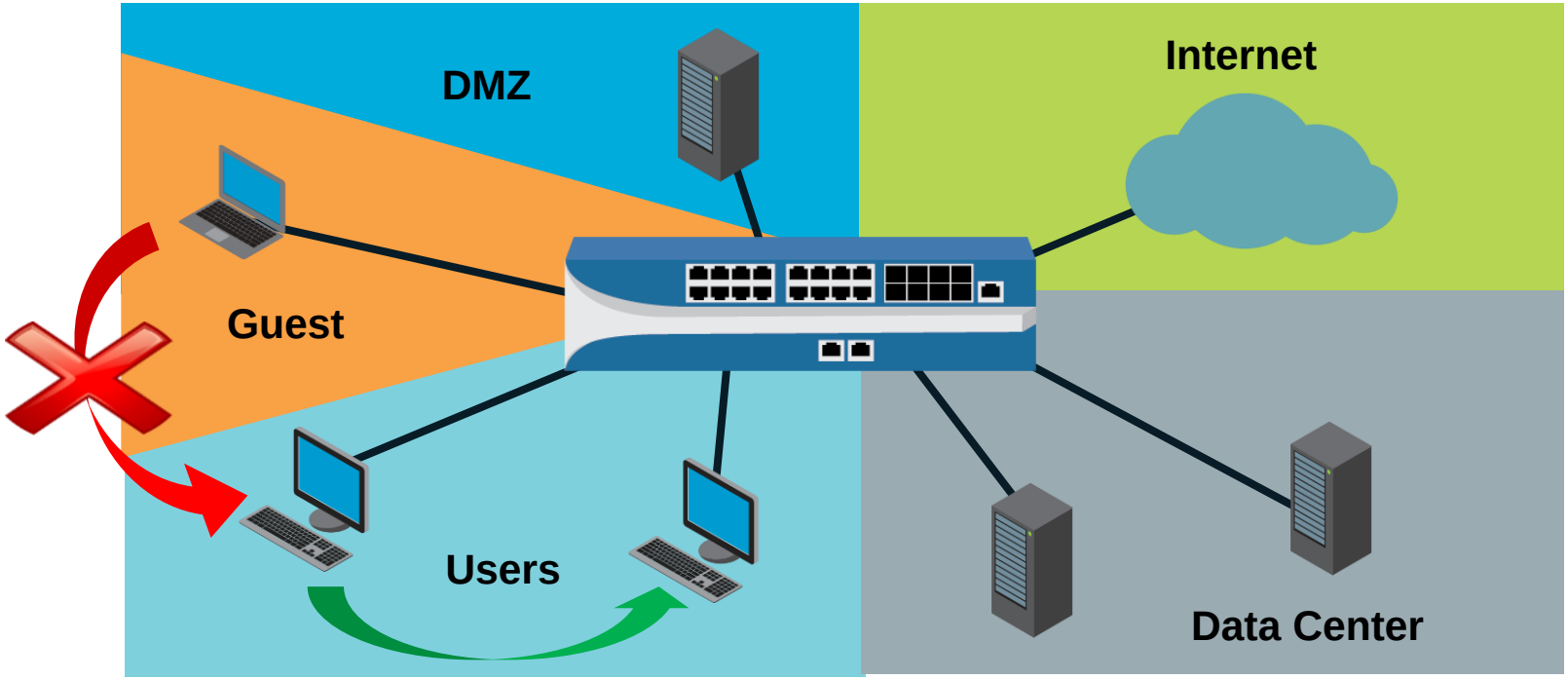
In-Band Network Interfaces

- Each interface is assigned to a single zone.
- A zone can include multiple physical or logical interfaces.



Security Zones and Security Policy Rules

- Traffic within a zone is *allowed* by default.
- Traffic between zones is *denied* by default.



Interface Types and Zone Types

- Most NGFWs support different zone types and interface types
- Each zone type supports specific interface types. E.g., PAN NGFW supports five zone and the interface types

Tap Zone

Tap interfaces

Layer 2 Zone

Layer 2 interfaces

Layer 3 Zone

- Layer 3 interfaces
- VLAN interfaces
- Loopback interfaces
- Tunnel interfaces

Tunnel Zone

No interfaces assigned

Virtual Wire Zone

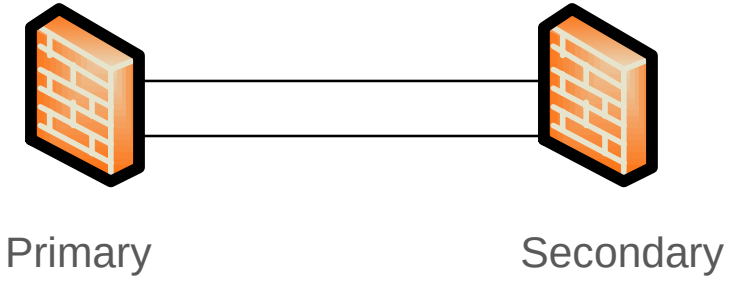
Virtual Wire interfaces

MGT and HA interfaces are not assigned to a zone

IP address required by these interfaces

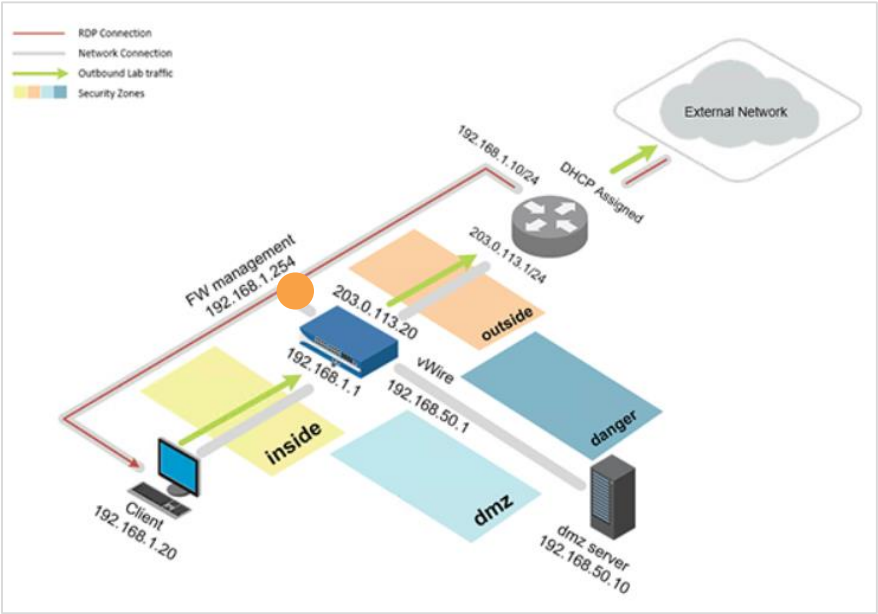
Interface Types and Zone Types

- Additional interface types are Management (MGT) and High Availability (HA).
- HA interfaces are not used to control normal network traffic; they are used for synchronization of a pair of firewalls deployed in a HA configuration.
- Because they do not control normal network traffic, they are not placed in a security zone.



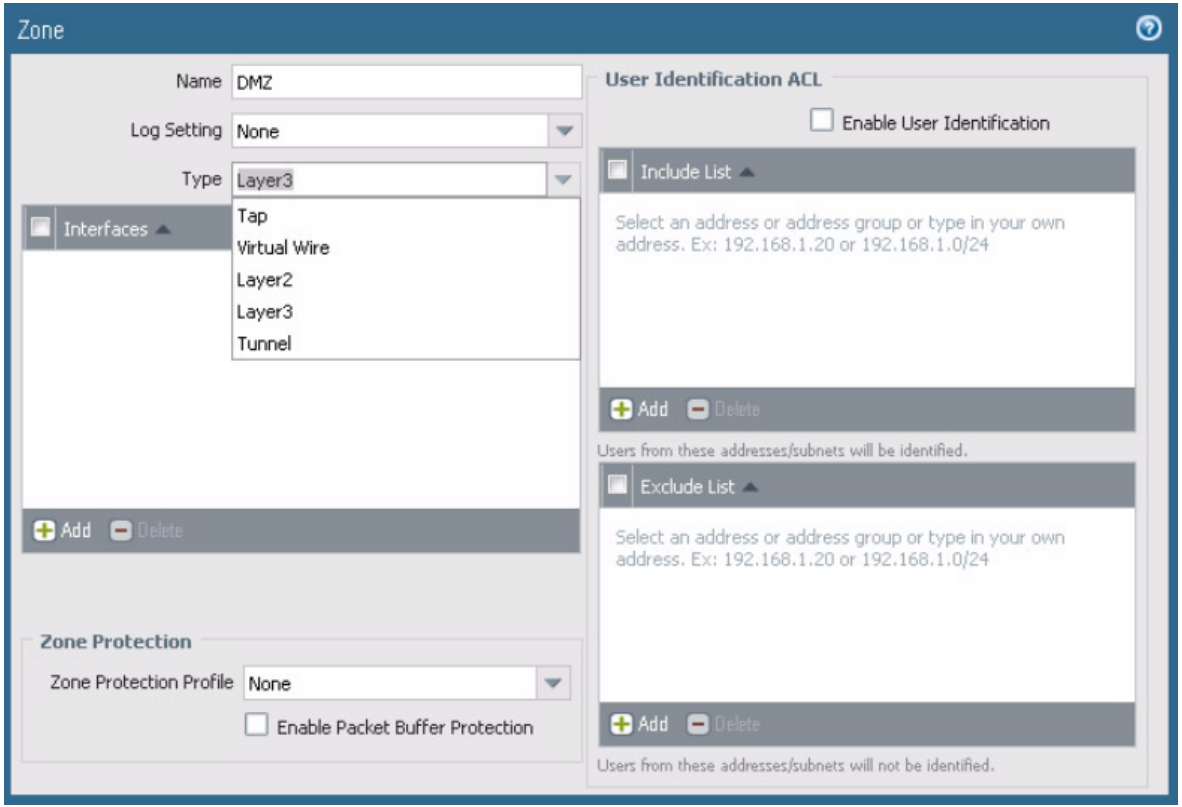
Interface Types and Zone Types

- The MGT interface is used for firewall management and is not assigned to a zone either.



Creating a Security Zone

Network > Zones > Add

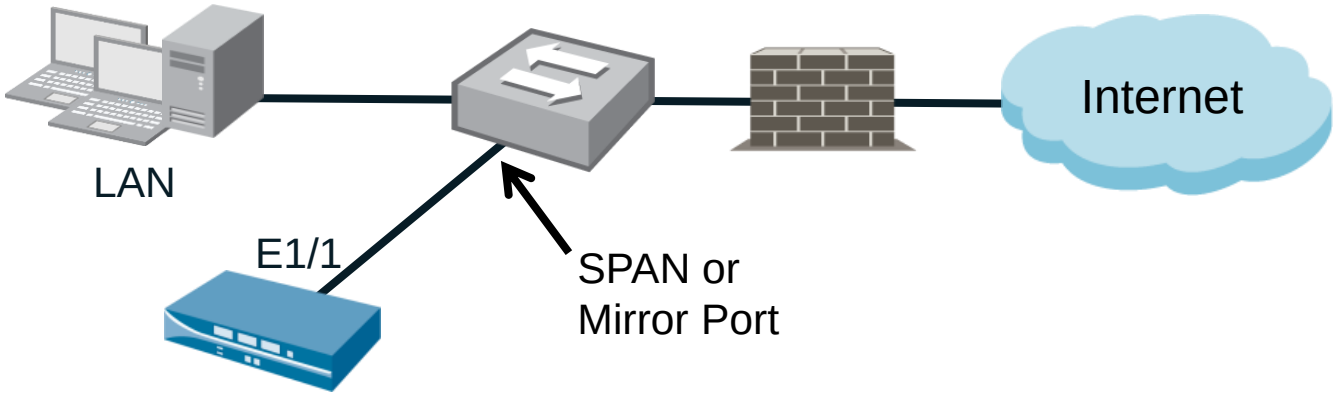


- Specify zone name
- Specify zone type
- Assign interfaces:
  - Must be appropriate type
  - Unassigned interfaces do not process traffic

Tap Interfaces

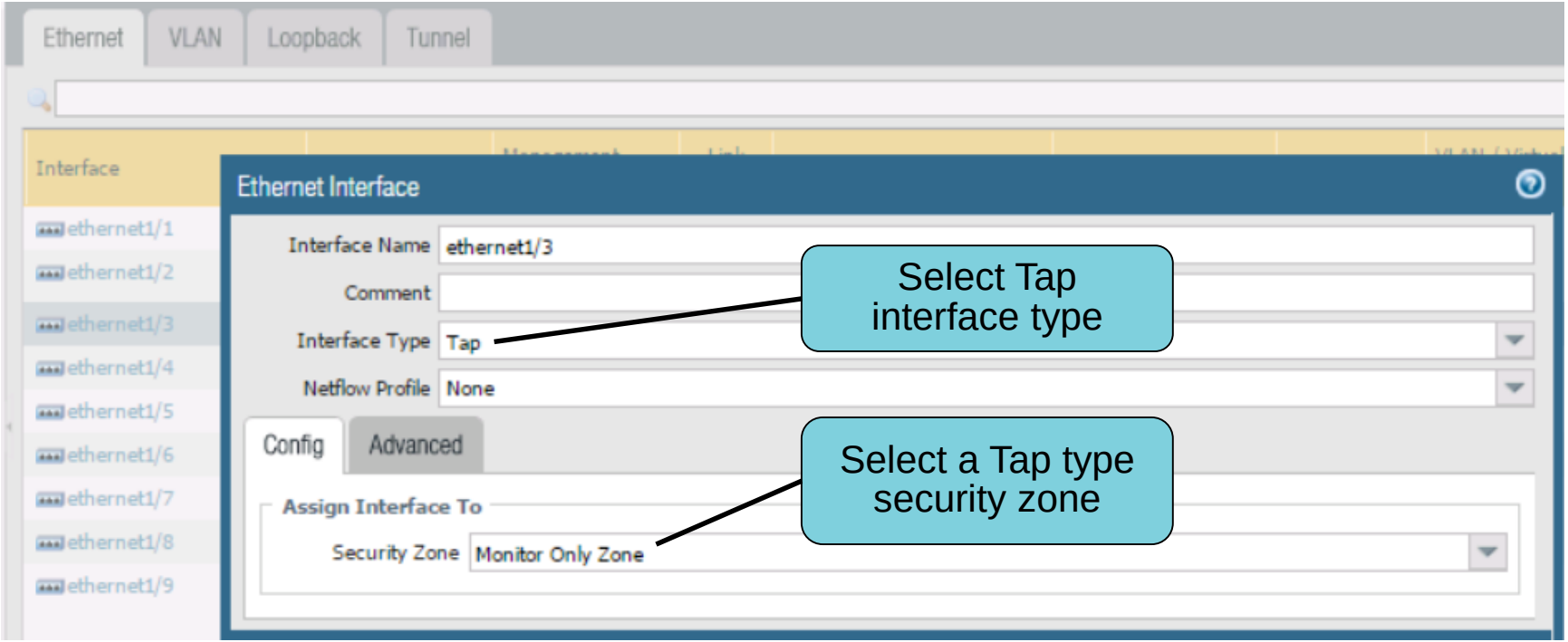
Tap Interface

- Most networks rely on tap interfaces for traffic monitoring
- Enables passive monitoring of switch traffic from a Switch Port Analyzer (SPAN) or mirror port
- Cannot control traffic or perform traffic shaping
- Must be assigned to a Tap zone
- Use Traffic log information to configure Security policy rules



Configuring a Tap Interface

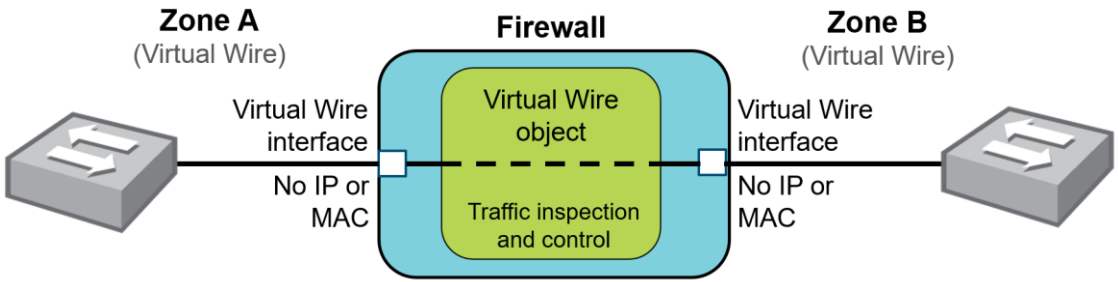
Network > Interfaces > Ethernet > <select_interface>



Virtual Wire Interfaces

Virtual Wire Interface

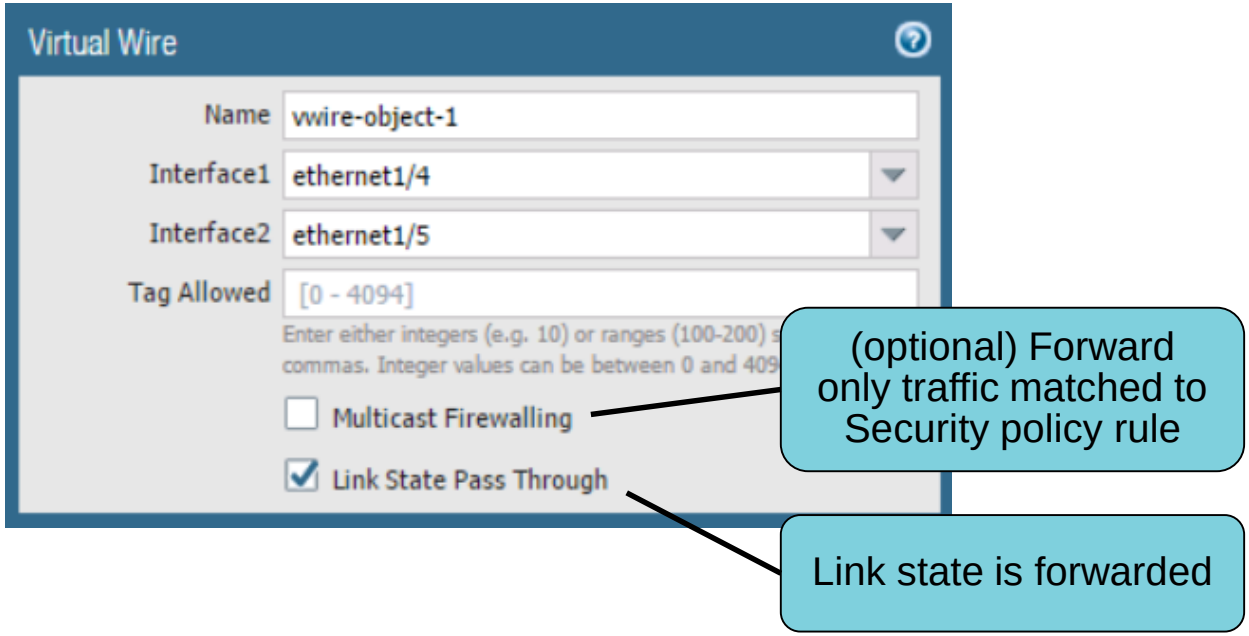
- A virtual wire binds two firewall interfaces together through virtual wire object
- Typically used when no switching or routing is needed
- No configuration changes for adjacent network devices
- NGFW can examine traffic, decrypt traffic, and more (e.g., application ID, content ID)



Configuring a Virtual Wire Object

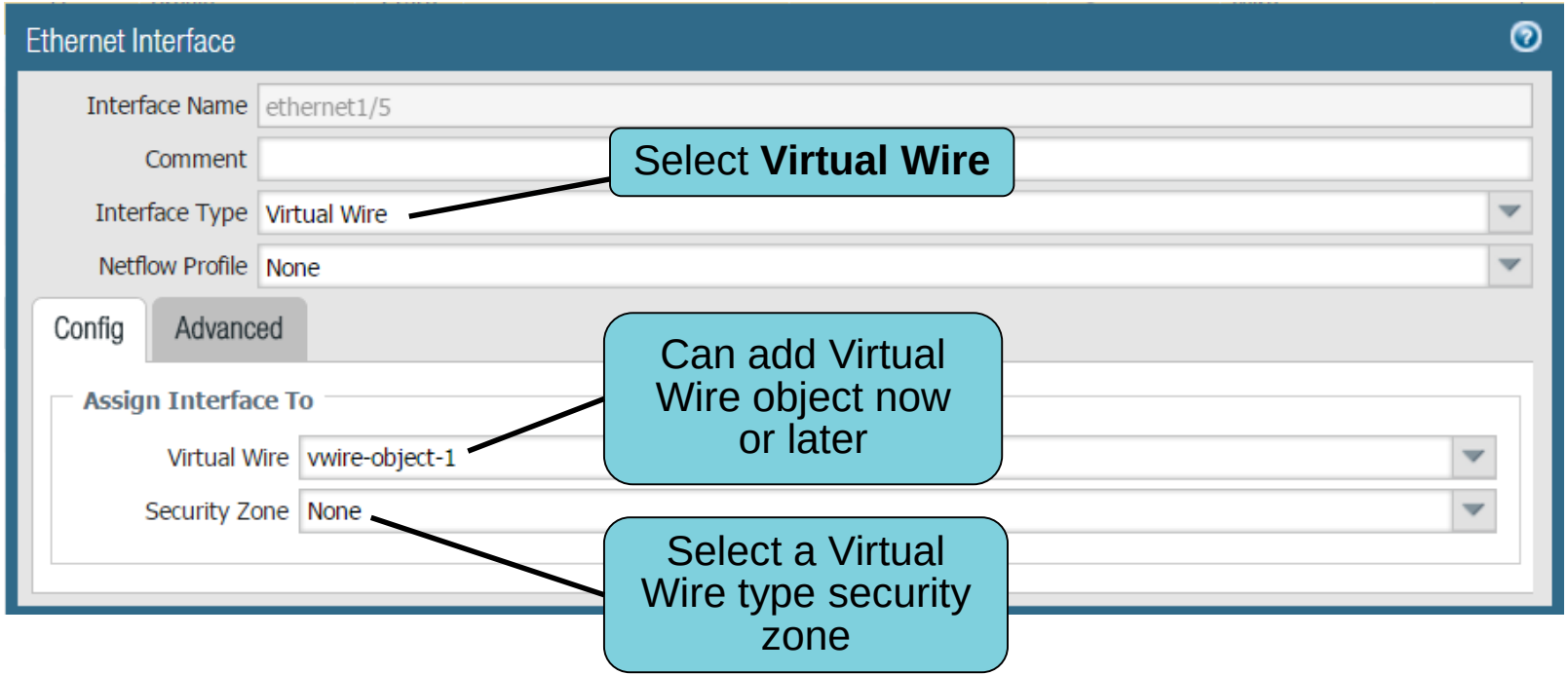
- Virtual Wire object connects to Virtual Wire interfaces.
- A virtual wire can accept traffic based on 802.1Q VLAN tags:
  - 0 = untagged traffic

Network > Virtual Wires > Add

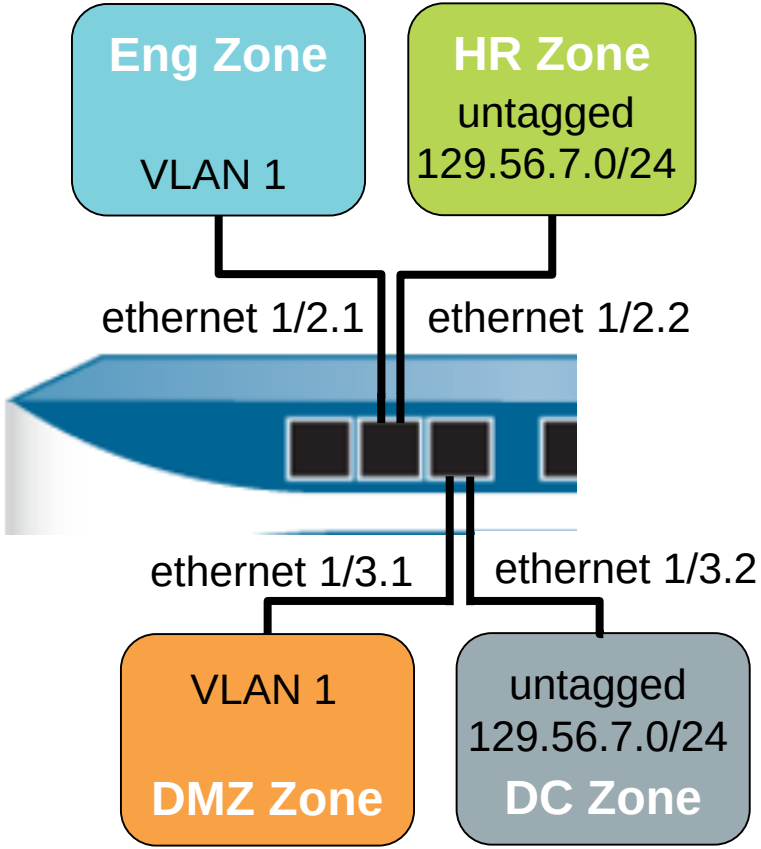


Configuring a Virtual Wire Interface

Network > Interfaces > Ethernet > <select_interface>



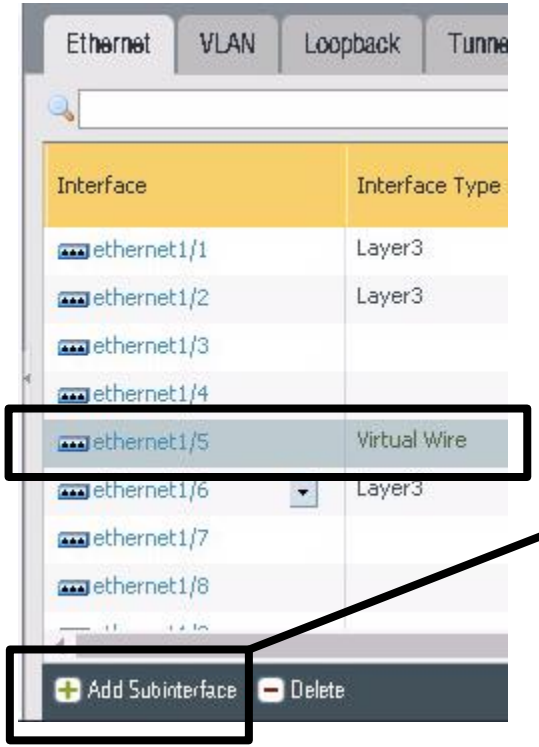
Virtual Wire Subinterfaces

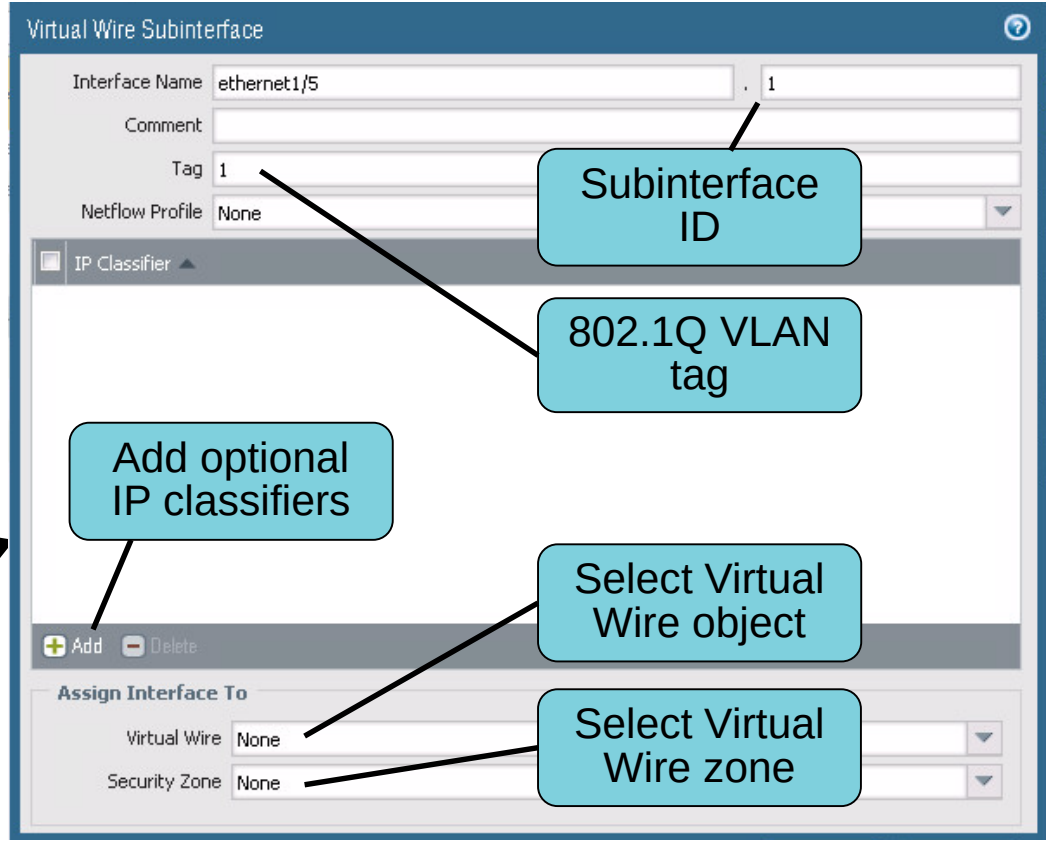


- Read and process traffic based on:
  - VLAN tags (1-4094)
  - VLAN tags and IP classifiers (source IP)
  - IP classifiers (untagged traffic, source IP)
- Common uses include:
  - More granular security rules
  - Logically splitting network traffic

Configuring a Virtual Wire Subinterface

Network > Interfaces > Ethernet

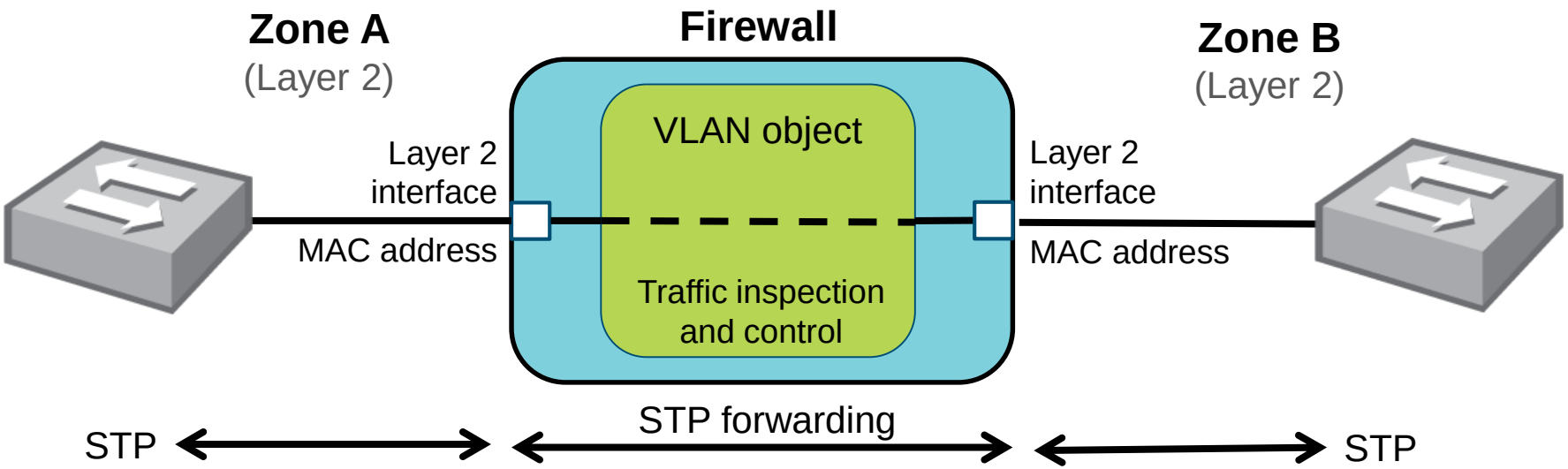




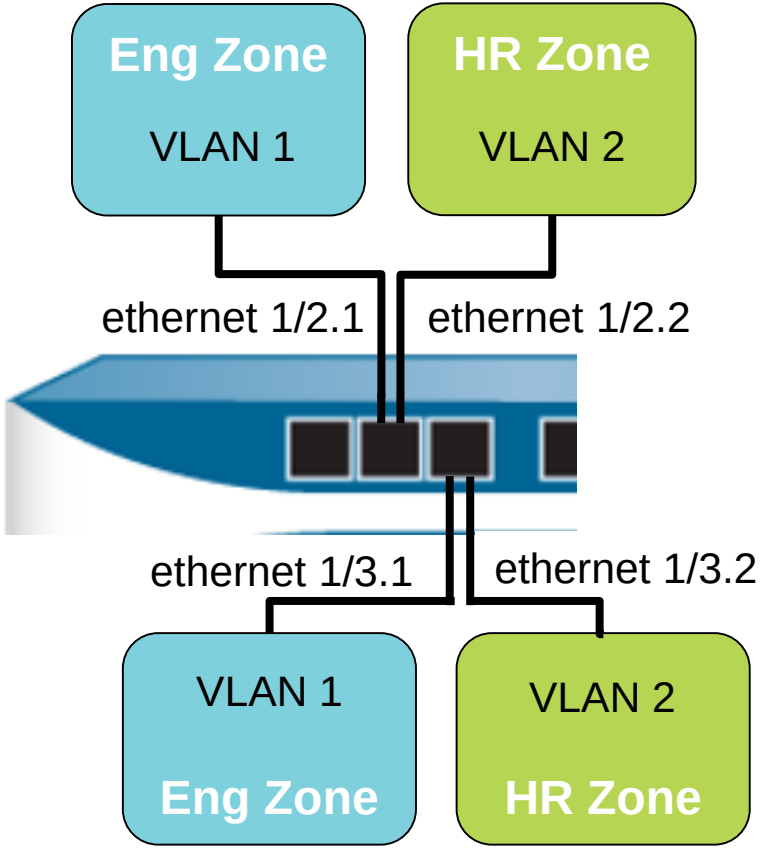
Layer 2 Interfaces

Layer 2 Interfaces

- Provides *switching* between two or more interfaces through VLAN object
- Typically used when no routing is needed
- No IP address is assigned to a layer 2 interface (no routing)



Layer 2 Subinterfaces

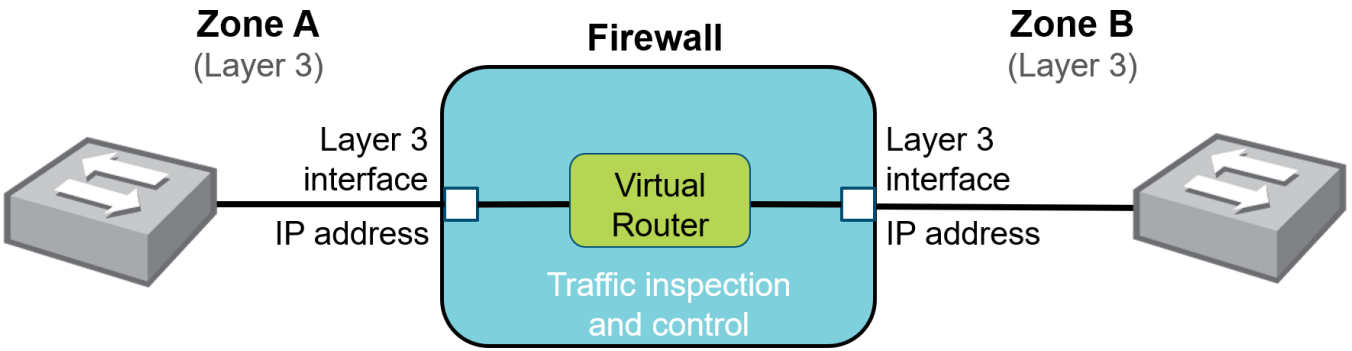


- Assign subinterfaces to zones
- VLAN traffic isolated by subinterfaces:
  - Need route between VLANs
  - Security policy blocks interzone traffic by default
- Useful configuration for multi-tenant networks

Layer 3 Interfaces

Layer 3 Interfaces

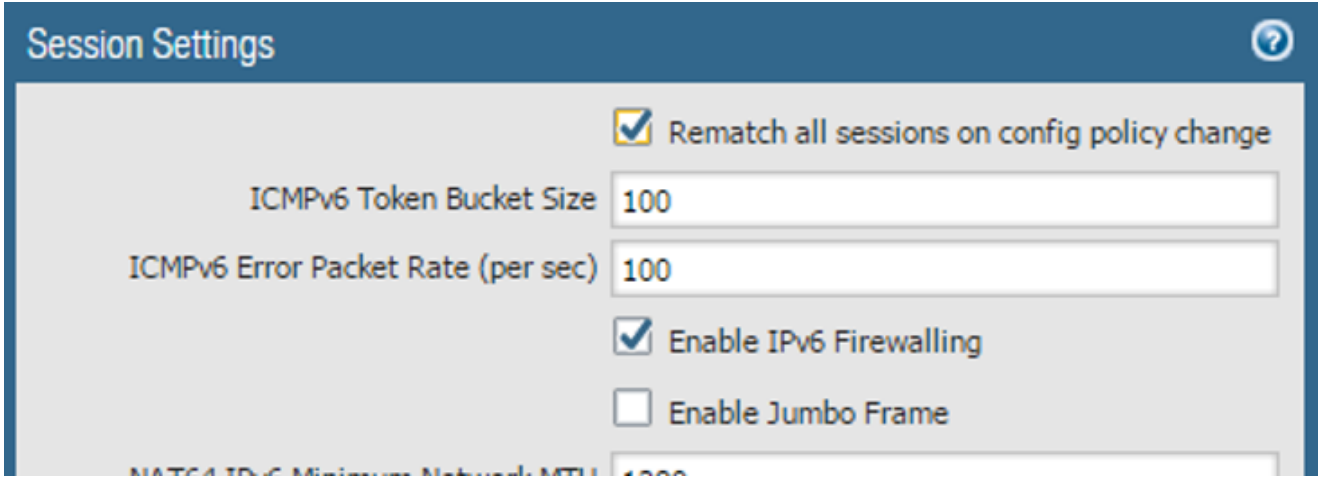
- Enables routing between multiple interfaces:
  - Requires a virtual router
- Each layer 3 interface must have an IP address
- All NGFW features are available at a Layer 3 interface: App-ID, Content-ID, User-ID, SSL decryption, NAT, and QoS



IPv4 and IPv6

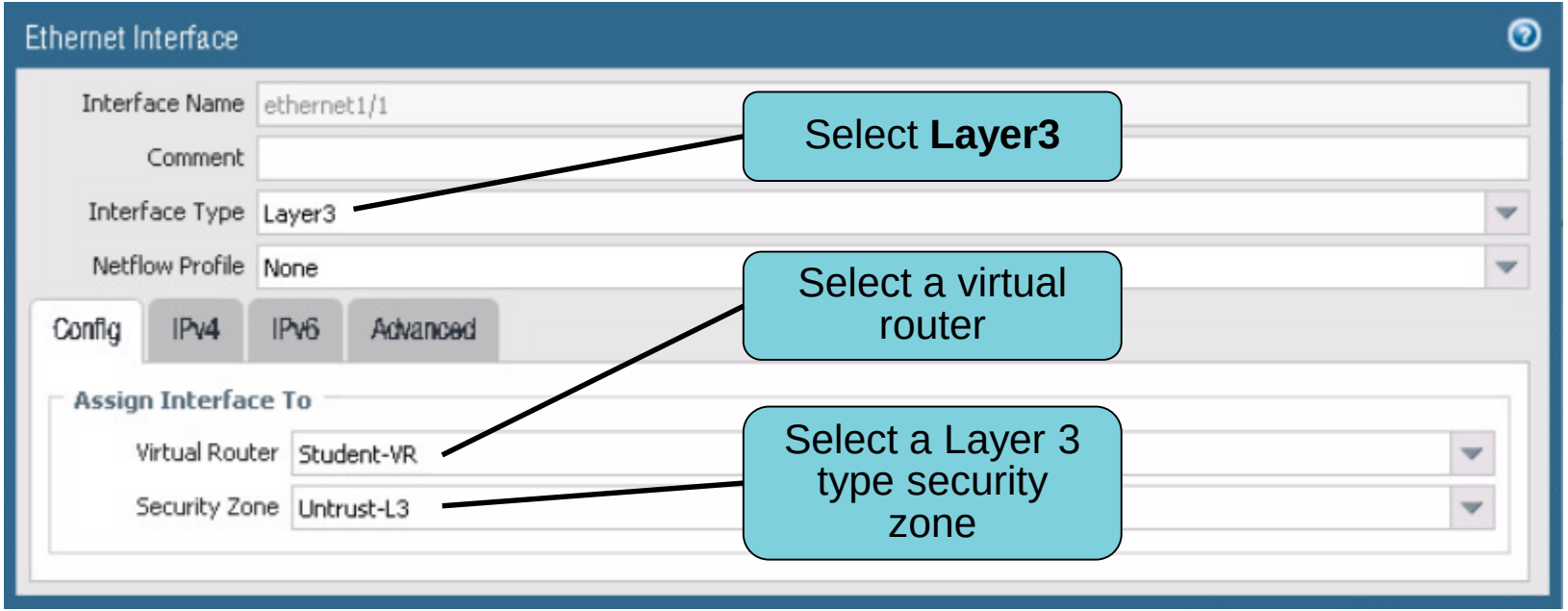
- Layer 3 interfaces support IPv4 and IPv6.
- To support IPv6 addresses, you must enable IPv6 on the firewall.

Device > Setup > Session > Session Settings



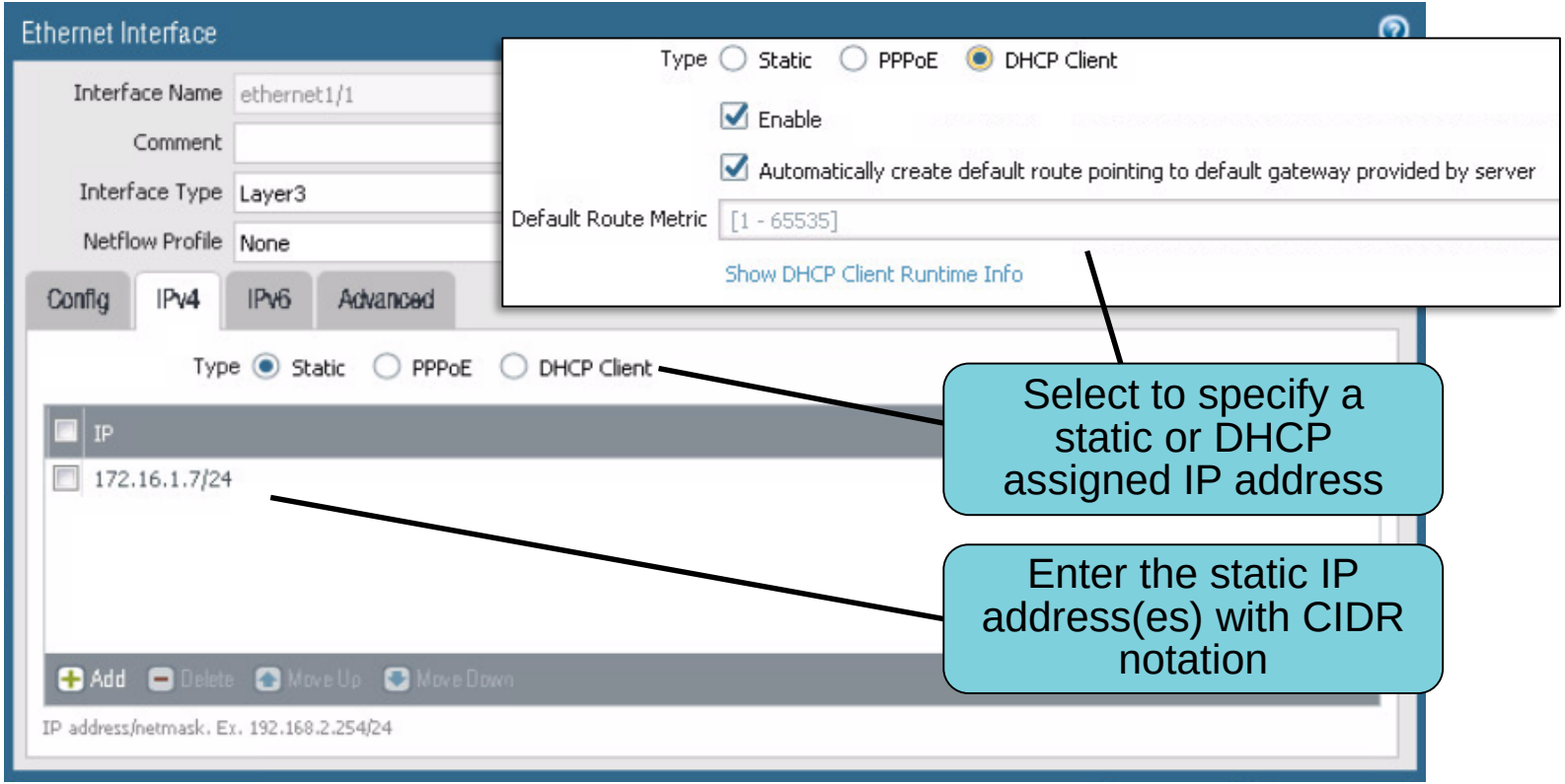
Configuring a Layer 3 Interface – Config

Network > Interfaces > Ethernet > <select_interface>



Configuring a Layer 3 Interface – IPv4

Network > Interfaces > Ethernet > <select_interface>



Configuring a Layer 3 Interface – IPv6

Network > Interfaces > Ethernet > <select_interface>

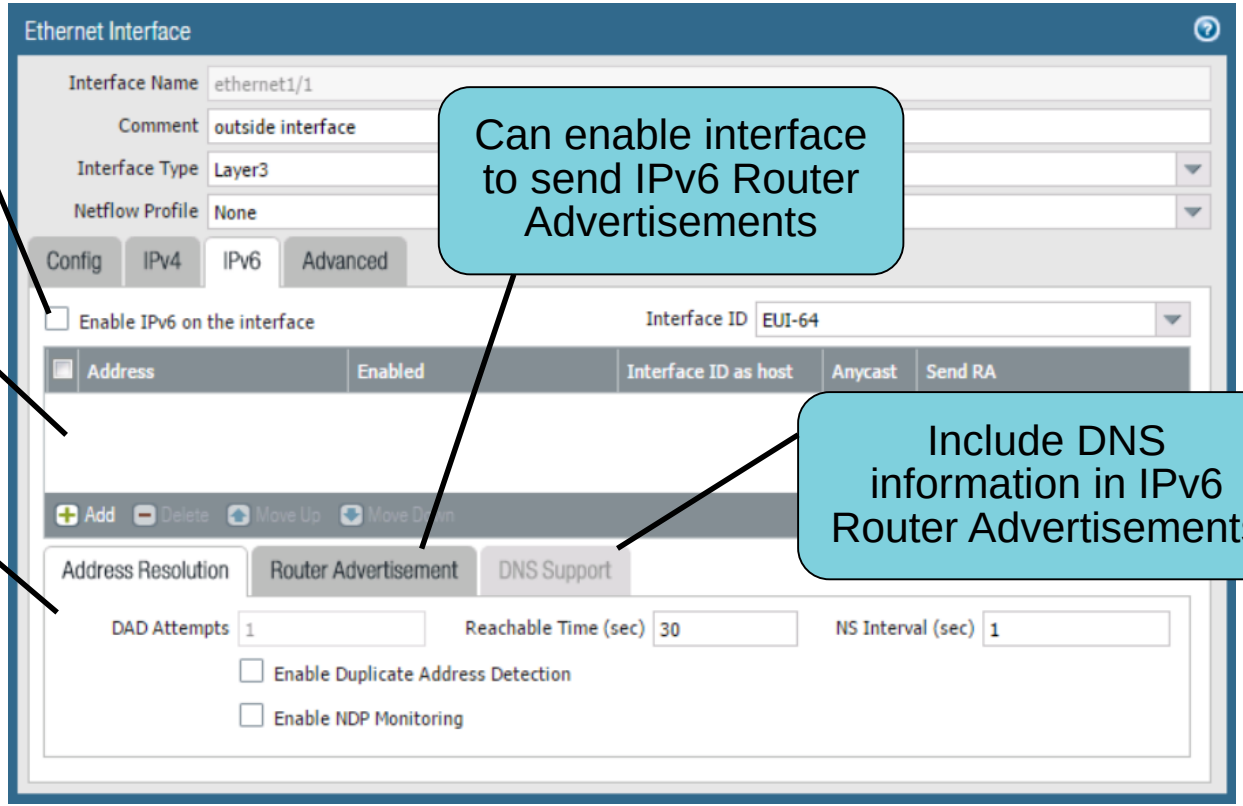
Enable link-local IP address

Can add Global addresses

Can enable Neighbor Solicitation to detect duplicate address assignments

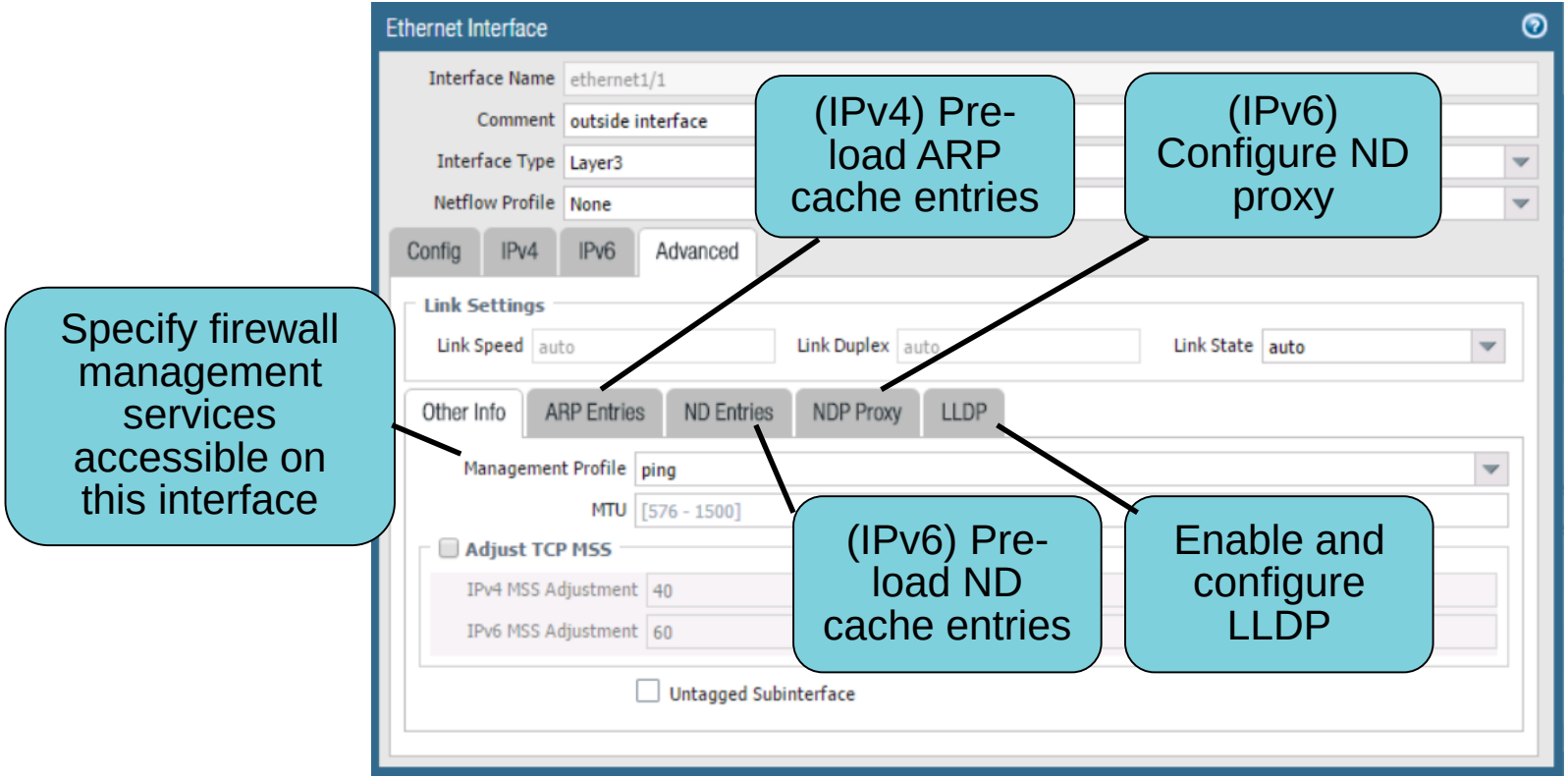
Can enable interface to send IPv6 Router Advertisements

Include DNS information in IPv6 Router Advertisements



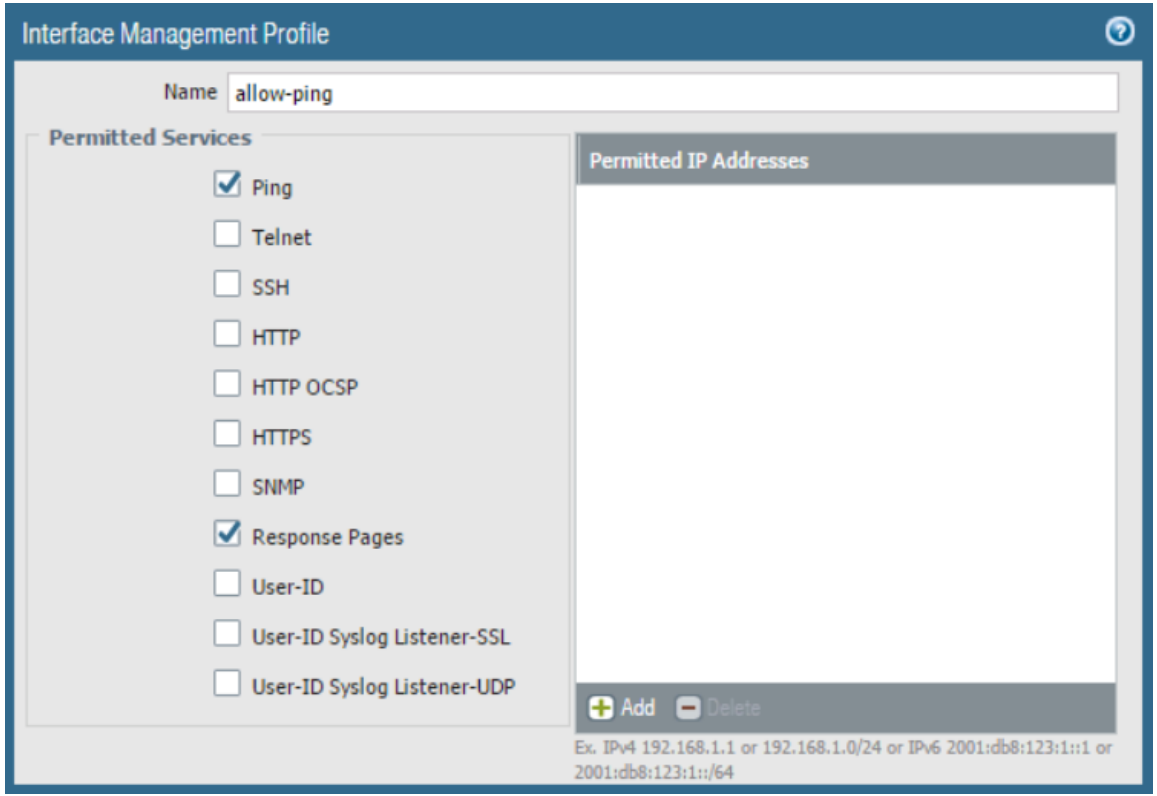
Configuring a Layer 3 Interface – Advanced

Network > Interfaces > Ethernet > <select_interface>



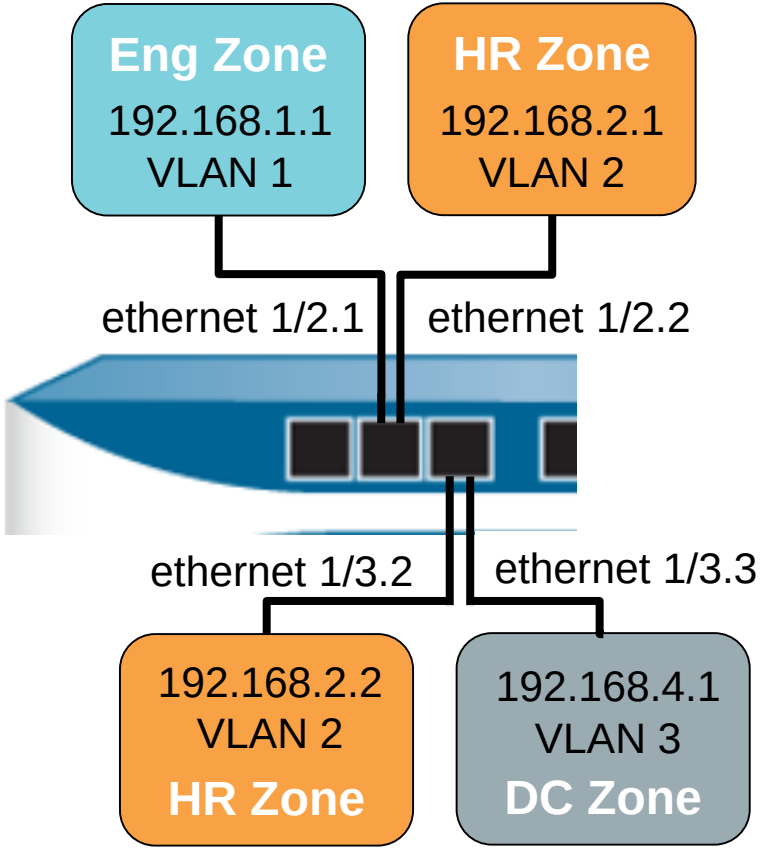
Interface Management Profile

Network > Network Profiles > Interface Mgmt > Add



- Defines which firewall management services are accessible from a traffic interface
- Can be applied to Layer 3, loopback, and tunnel interfaces

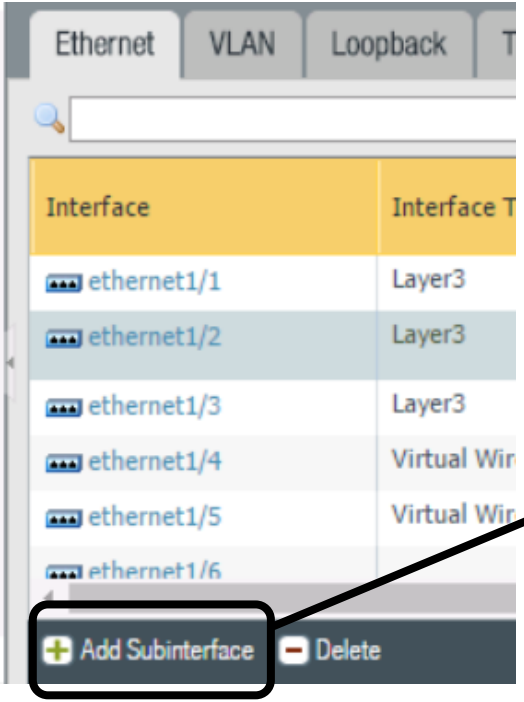
Layer 3 Subinterfaces

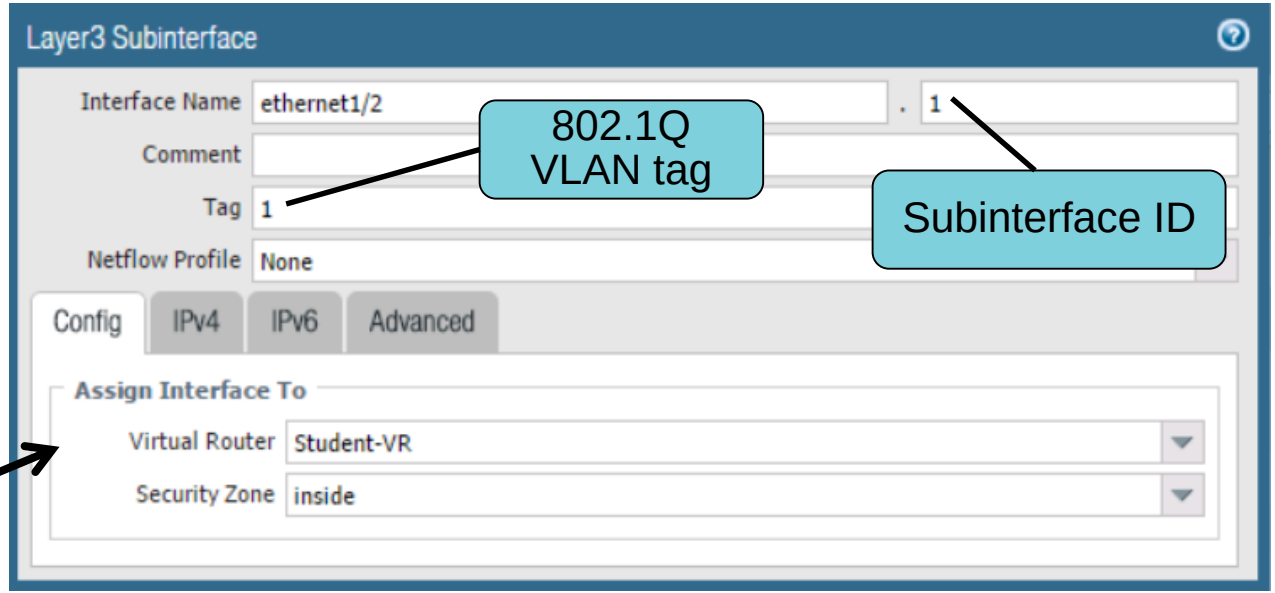


- Assign subinterfaces to zones
- Traffic in each VLAN is isolated:
  - Need virtual router to connect VLANs
  - Security policy blocks interzone traffic by default
- Useful configuration for multi-tenant networks

Configuring a Layer 3 Subinterface

Network > Interfaces > Ethernet



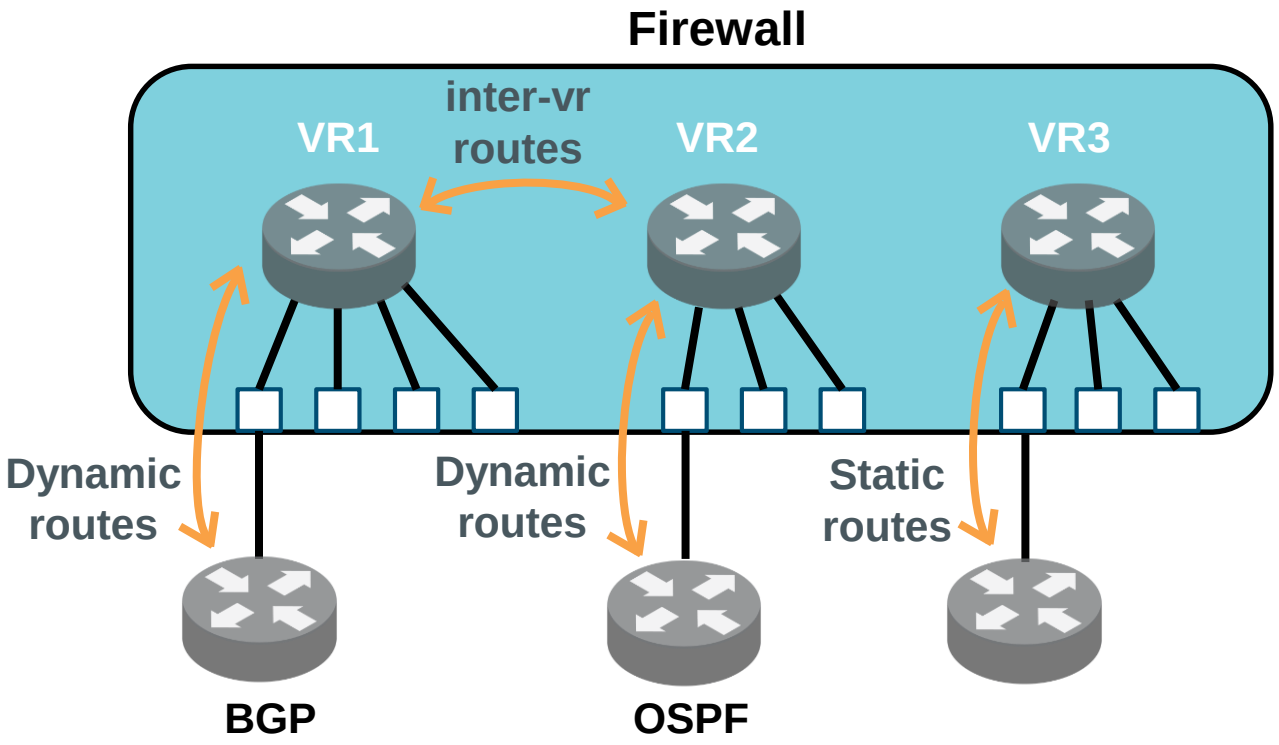


Configure remaining options as normal Layer 3 interface

Virtual Routers

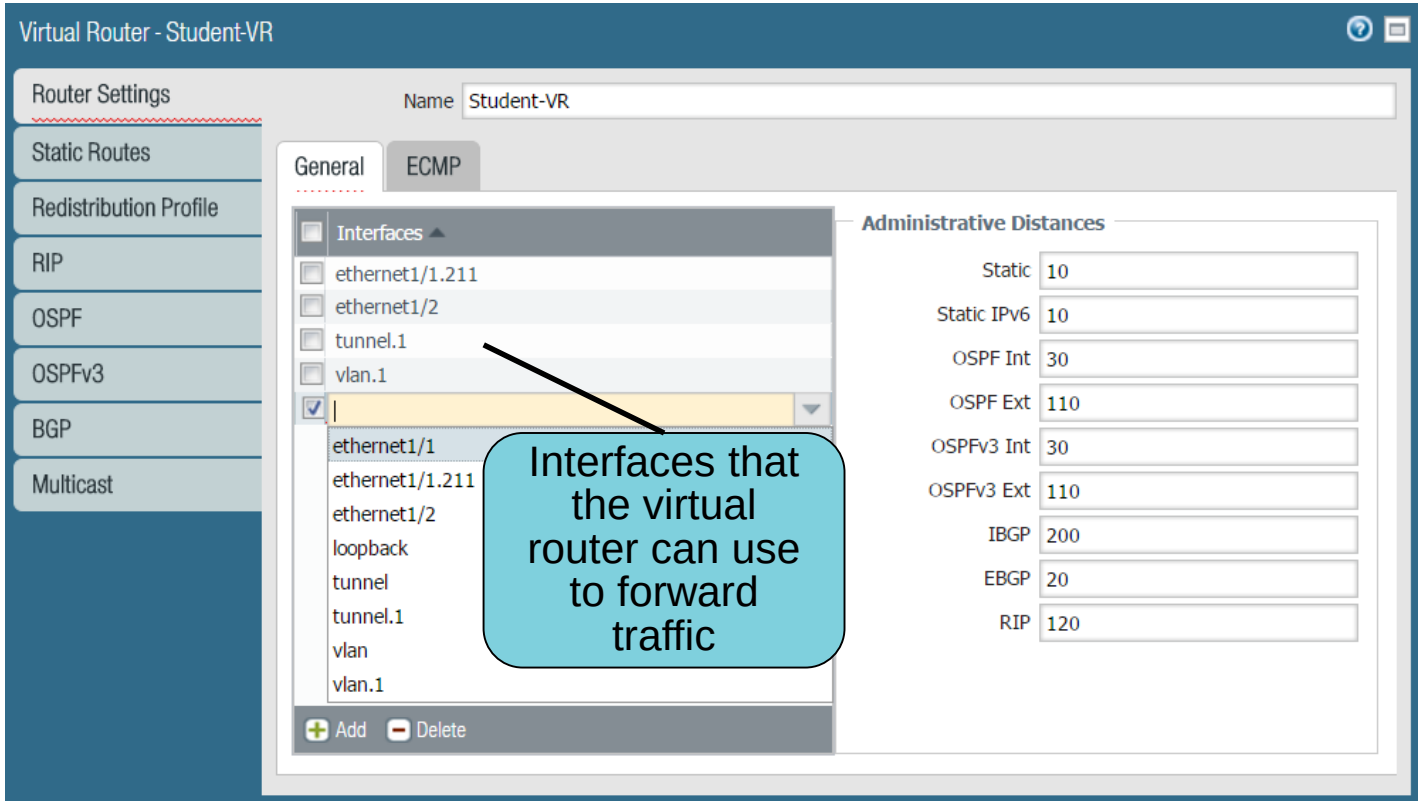
Virtual Routers

- Support one or more static routes
- Support dynamic routing:
  - RIPv2
  - OSPFv2
  - OSPFv3
  - BGPv4
- Support multicast routing
  - PIM-SM
  - PIM-SSM



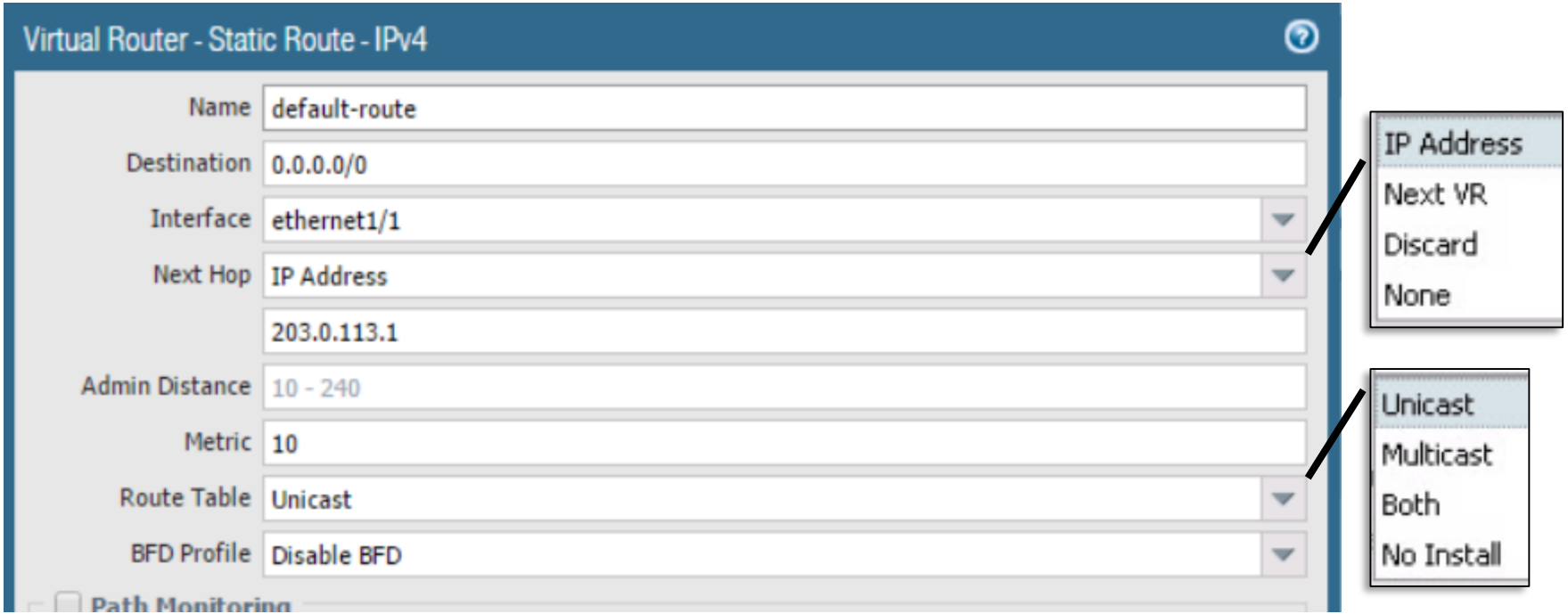
Virtual Router General Settings

Network > Virtual Routers

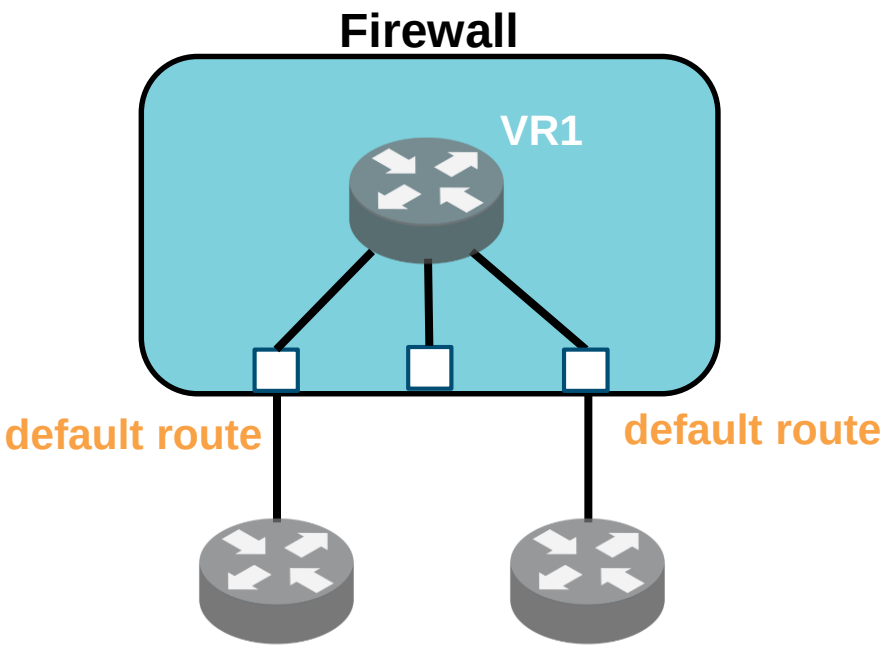


Adding a Static Default Route

Network > Virtual Routers > Static Routes > Add



Multiple Static Default Routes



- Can configure multiple default routes
- Route with lowest metric is used
- Path monitoring determines if routes are usable
- Firewall switches default route during path failure
- Supports failback

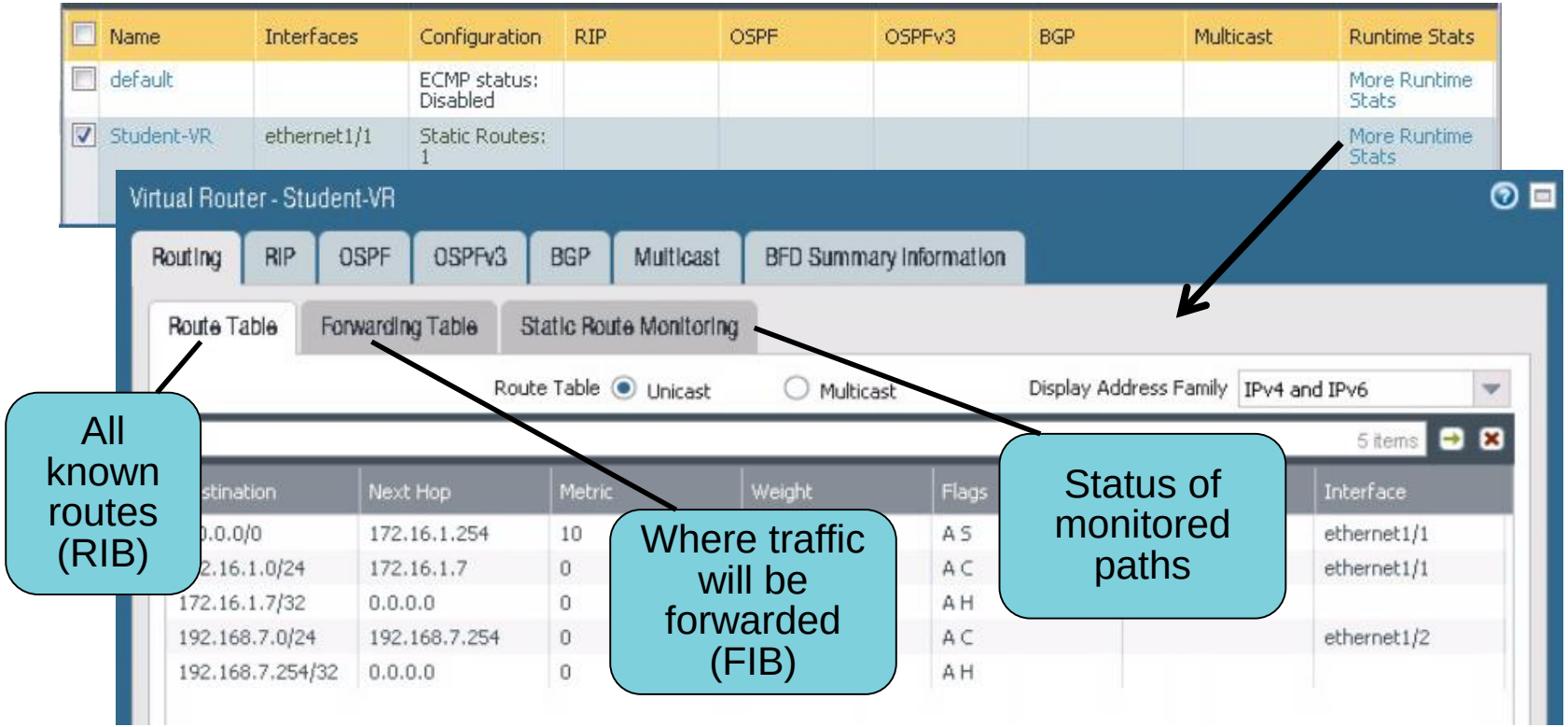
Static Route Path Monitoring

Network > Virtual Routers > Static Routes > Add

- Uses ping to test reachability to stable upstream devices
- Testing continues after failure
- Will remove or re-add static routes

Troubleshooting Routing

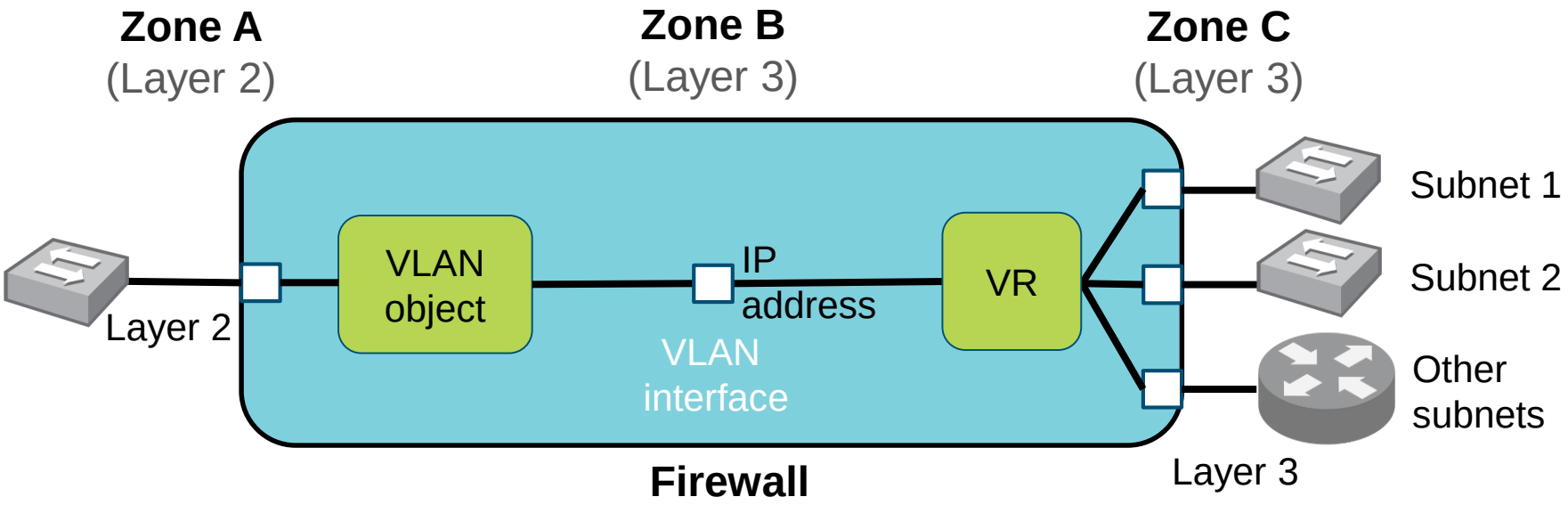
Network > Virtual Routers



VLAN Interfaces

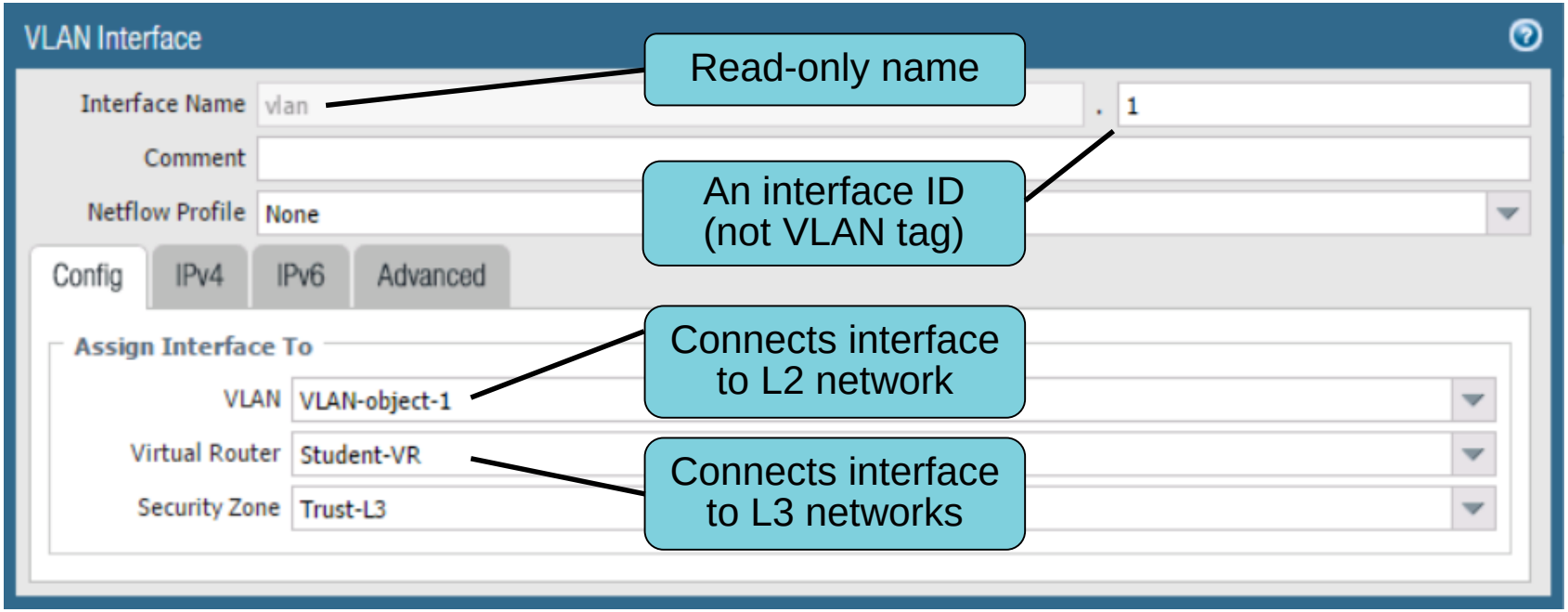
VLAN Interfaces

- Are assigned an IP address
- Connect Layer 2 to Layer 3



Configuring a VLAN Interface

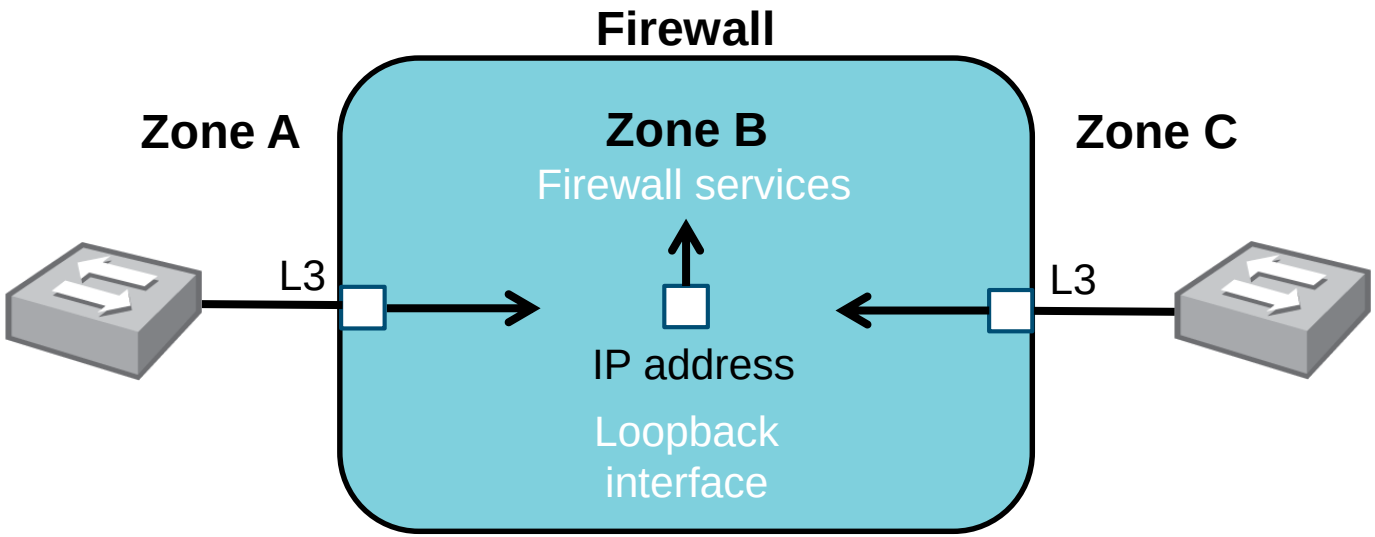
Network > Interfaces > VLAN > Add



Loopback Interfaces

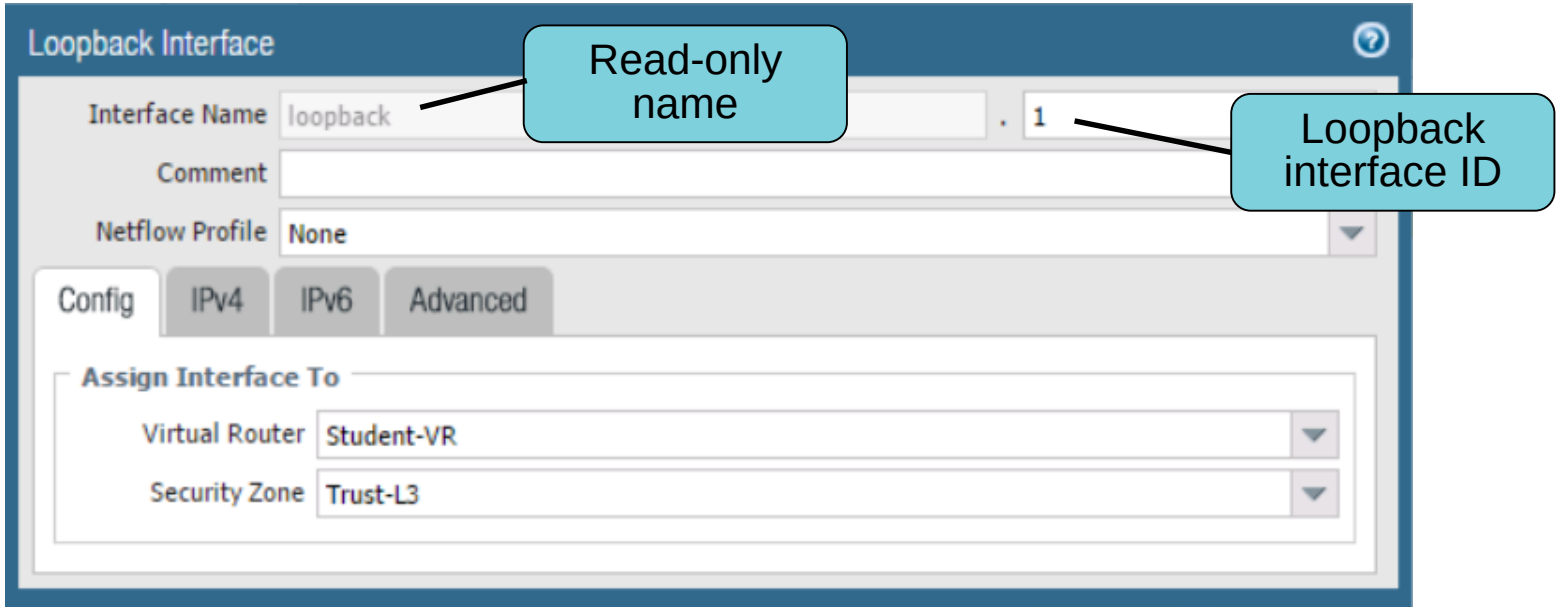
Loopback Interface

- Logical interface with an IP address
- Behaves like a host interface
- Used to provide access to firewall services



Configuring a Loopback Interface

Network > Interfaces > Loopback > Add

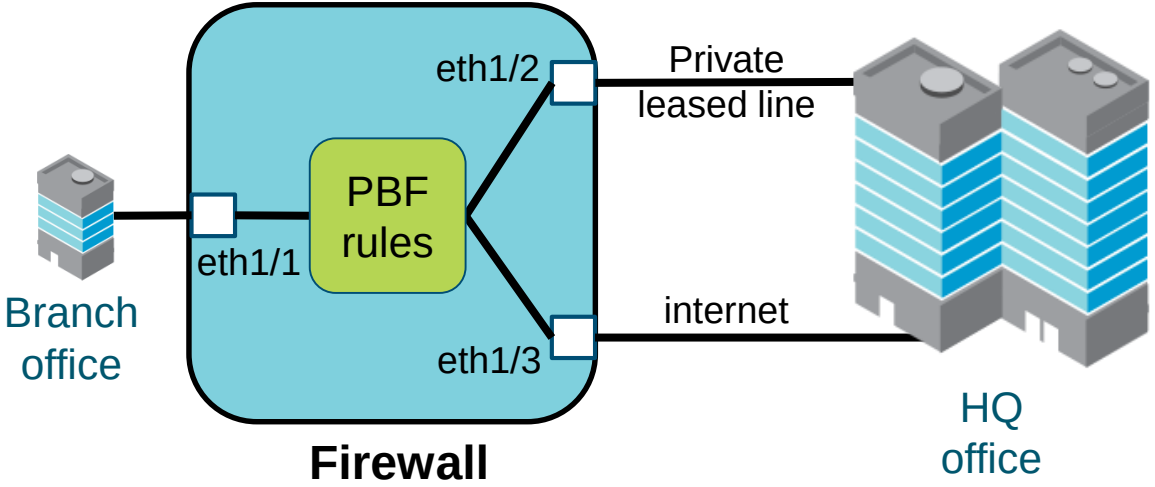


Do not assign a netmask to the IP addresses.

Policy-Based Forwarding

Policy-Based Forwarding

- Specifies different egress interface from that specified in route table
- Possible use for performance or security reasons



<u>Specify egress interface for:</u>

- Bandwidth sensitive applications
- Unencrypted applications

<u>Specify egress interface for:</u>

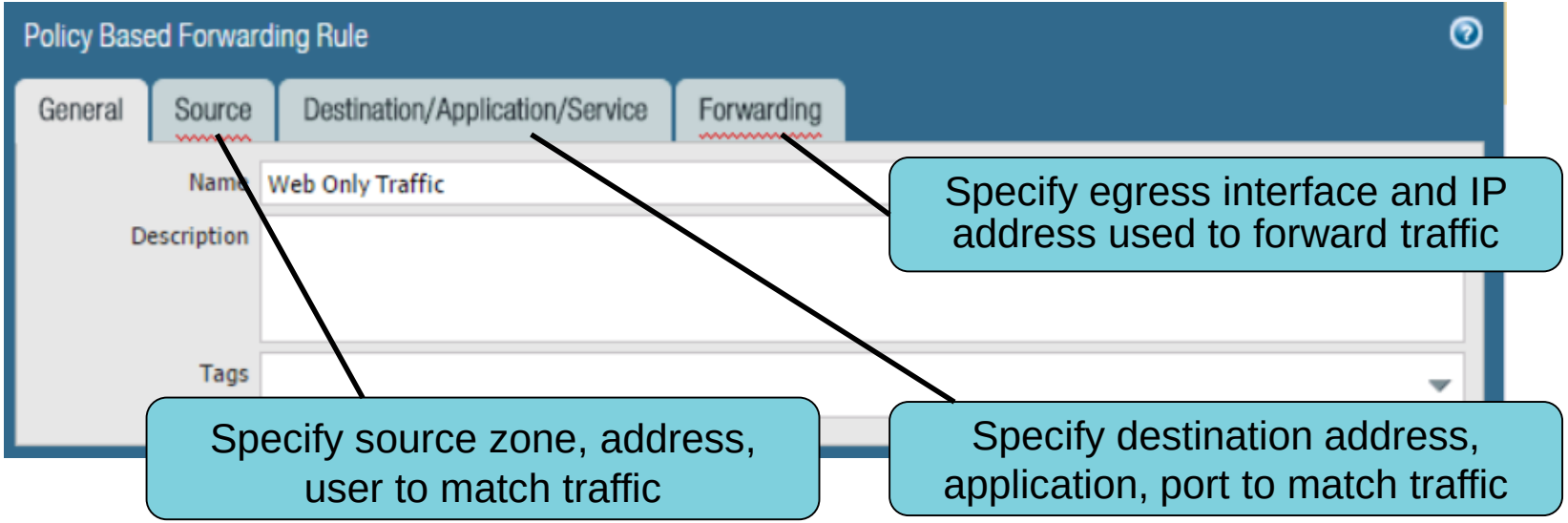
- Non-bandwidth sensitive applications
- Encrypted applications

PBF Rules

- PBF rules use match criteria to match traffic.
- PBF path monitoring enables the firewall to verify network path connectivity.

Policies > Policy Based Forwarding

			Source			Destination				Forwarding			
	Name	Tags	Zone/Interface	Address	User	Address	Application	Service	Action	Egress I/F	Next Hop	Enforce Symmetric Return	Profile
1	Mail OnlyTraffic	none	Trust-L3	any	any	Mail Servers	any	SMTP	forward	ethernet1/6	201.34.17.6	false	default

Configuring PBF – General Tab

Policies > Policy Based Forwarding > Add